

**Treatment Success and Failure Prediction Using Machine Learning Models
for Pulmonary Tuberculosis Patients Under the Department of Health-Cordillera
Administrative Region (DOH-CAR) Tuberculosis-Directly Observed Treatment,
Short-course (TB-DOTS) Program**

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Abstract

Pulmonary Tuberculosis (PTB) still lingers as a big public health concern in the Cordilleras, where differences between patients and gaps in the treatment program still contribute to treatment failure. This study develops and evaluates machine learning models to predict treatment success or failure among patients in DOH-CAR TB-DOTS facilities for the years from 2015 to 2025. A preprocessing pipeline performs the following: schema harmonization, patient pseudonymization (for anonymization), and a diagnostic-first missing data handling protocol. The protocol classifies each variable's missingness mechanism as Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR), then routes the variable to one of four strategies: column exclusion, listwise deletion, missing indicator imputation, or stochastic Multivariate Imputation by Chained Equations (MICE). Supervised tree-based classifiers (XGBoost, LightGBM, and Random Forest) are trained alongside Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) temporal architectures to model longitudinal treatment dynamics over the 6- to 12-month treatment course. Because clinical decision support needs more than an accurate risk score, Shapley Additive Explanations (SHAP) are used to show which patient-level factors contributed to each prediction so that clinicians can review the system's outputs. The validated models are embedded in a Clinical Decision Support System (CDSS) that provides dynamic risk estimates, SHAP-grounded factor explanations, and bounded narrative summaries. The CDSS is intended to support early clinical intervention while preserving clinician judgment.

Keywords: tuberculosis, machine learning, treatment outcome prediction, clinical decision support system, explainable AI

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Pulmonary Tuberculosis (PTB) is a chronic infectious disease caused by *Mycobacterium tuberculosis* (MTB), which is a slow-growing, acid-fast bacillus. And because the disease progresses gradually and that treatment can fail, mitigation is difficult. Effective treatments exist, but PTB is still widespread. It tracks closely with poverty, which is naturally followed by undernutrition, and comorbidities as well such as diabetes and HIV, all of which weaken patients' immune defenses and raise susceptibility to the disease. These conditions also delay the diagnosis and complicate treatment. TB is one of the leading causes of death worldwide, and in 2018 alone, there were around 10 million new cases and 1.5 million deaths, and about half a million of those were resistant to rifampicin.

In the Philippines, TB still remains as a big problem with regards to public health, and the World Health Organization (WHO) lists the country as one of the biggest contributors to the global TB burden, estimated at around 6.8 percent of cases coming from the Philippines. Before the COVID-19 pandemic, Philippine TB rates were also among the highest in Asia: over 554 cases per 100,000 people in 2019. In 2023, about 37,000 Filipinos died of TB out of 739,000 diagnosed cases, and the WHO classifies the country as a high-burden setting for both MDR-TB and TB/HIV co-infection. That same year the DOH logged 575,770 TB cases in its Integrated TB Information System. The number itself signals heavy under-registration and underdiagnosis. For treatment, the DOH follows CDC-recommended regimens for active and latent TB under the TB-DOTS strategy, widened in 2003 to take in private clinics, pharmacies, and hospitals. Gaps persist even so, in diagnosis, in treatment monitoring, in patient support, and most of all in managing MDR-TB.

Baguio City shows the same gap up close. In 2022 the DOH-CAR confirmed 437 TB cases there but counted more than 4,405 that were missing or never reported, which puts the city near the top for surveillance and treatment work. Passive case-finding, stigma, and the cost of care drive

that gap, and the undetected patients keep spreading the disease in their communities .

The large population and a lot of daily traffic (both foot and vehicular) in the city may increase opportunities for tuberculosis to spread. The colder climate can also create more ideal conditions for the transmission of *Mycobacterium tuberculosis*. The city also has a more inflated local load of cases because it takes in more complex tuberculosis cases from neighboring provinces because of it being a tertiary referral center . Patient-level factors also play an important role in treatment outcomes such as comorbidities like diabetes mellitus, HIV infection, and malnutrition, all of which weaken their immune response and may reduce treatment completion. Food insecurity, poor housing conditions, and transportation costs can also interrupt adherence to directly observed treatment short-course (DOTS) .

All in all, these constraints argue for strategies that work on more than one facet of assessment. Treatment failure not only worsens the outcome for the singular patient, it also extends the period for which transmission can happen in the community and in-turn speeds up the emergence of multidrug-resistant tuberculosis (MDR-TB), defined as resistance to at least rifampicin and isoniazid. MDR-TB requires regimens that are longer, more toxic, and more expensive, and it carries much lower success rates. The result is a wider threat to regional disease control in the Philippines .

Risk Factors for Treatment Failure

To ensure consistent evaluation of tuberculosis control efforts, the Department of Health (DOH) established standardized criteria for classifying treatment outcomes through the National Tuberculosis Program (NTP). These outcome definitions are important for monitoring program performance, comparing findings across studies, and identifying factors associated with treatment outcomes. Treatment success and failure do not occur randomly, as they are often influenced by identifiable patient, clinical, and program-related risk factors. Mortality remains the primary outcome examined in most studies on treatment failure, although some also include disease pro-

gression, multidrug-resistant tuberculosis, and loss to follow-up as indicators of poor treatment outcomes .

One of the risk factors is close or direct contact with patients who have active TB infection where patients who acquired the infection through close contact were at higher risk of delayed diagnosis and more likely to be non-adherent, either by refusing or quitting the regimen . Another risk factor is recurrence of the infection or a history of receiving TB treatment because patients who missed follow-ups had a higher risk of reactivation and mortality, and were harder to treat than patients with no prior history of TB . Frequent missed or interrupted treatment also leads to drug resistance .

Comorbidities have consistently shown a significant influence on tuberculosis treatment outcomes across different settings. In Eastern Ethiopia, HIV co-infection was a substantial risk factor for treatment failure because the weakened immune system makes treatment less effective . TB and HIV drug interactions also accelerate each other's progression and increase the occurrence of side effects, which can compromise treatment success . A study in Davao City, found a higher risk of treatment failure in comorbid patients: 58.82% of the 51 patients who experienced treatment failure had at least one comorbidity . Another major factor is Diabetes Mellitus (DM). TB patients with diagnosed DM have a higher risk of treatment failure, relapse, and death than non-diabetics . In a systematic review and meta-analysis, DM patients had a higher risk of treatment failure for Drug-resistant (DR) TB and mortality for MDR-TB, with odds ratios of 1.56 and 1.57 respectively . Another study reported that patients with both PTB and chronic conditions, especially DM, had reduced rates of sputum conversion and a higher probability of poor treatment outcomes, including death and progression to MDR-TB .

Taken together, these results say that no single factor drives TB treatment failure. It grows out of the way comorbidities, programmatic gaps, and socioeconomic strain feed into each other, and a one-factor analysis cannot see that. Catching it calls for a machine learning framework able to model the interactions all at once, the kind of model that makes earlier, better-aimed clinical intervention possible .

Machine Learning for TB Outcome Prediction

Previous studies on tuberculosis treatment outcomes have focused on sociodemographic characteristics, comorbidities, medical history, and other patient-level risk factors. The findings remain inconsistent and fragmented because methodological constraints limit how these factors are analyzed together. Most studies rely on traditional statistical analyses such as statistical tests or regression models, which examine one factor at a time, or a limited set of factors, while assuming linear relationships among variables . These approaches provide important evidence on individual risk factors, but they may miss the combined and non-linear effects that shape treatment success or failure.

This is where machine learning (ML) pulls ahead of conventional statistics: an algorithm that learns from high-dimensional data can pick up the subtle dependencies that decide whether treatment holds or fails. The variables in play, demographic, clinical, diagnostic, treatment-related, interact in complex, non-linear ways, the exact patterns a single-factor method tends to miss. Plenty of algorithms have been tried on the problem, logistic regression, Support Vector Machines (SVM), ensemble methods like Extreme Gradient Boosting (XGBoost) . Within TB, earlier work has aimed ML at predicting drug resistance, flagging high-risk patients, and tuning treatment strategy . The lesson across all of it is plain: a predictive model can carry risk-factor analysis further by reading several dimensions of patient data at once.

A clinical decision support system has to be interpretable. An ML model that hands back only an end result, with no account of why, is of little use at the bedside: a healthcare worker needs to understand the basis of a risk alert before acting on it. Tell TB-DOTS staff a patient is high-risk without naming the factors behind it, and they have little to trust or act on. The study answers this with Shapley Additive Explanations (SHAP), a post-hoc technique built on Shapley values that quantifies how much each variable contributed to a prediction and rolls up into both patient-level explanations and overall feature importance . It suits tree-based clinical models like XGBoost, Random Forest, and LightGBM, where accuracy can be high but the internal decision path is hard to inspect. In the healthcare ML literature, SHAP is among the most used explanation

methods, tabular data especially . Work on clinical ML explainability has leaned on it to surface real contributors to treatment failure: drug resistance, TB history, lab markers, and more .

Fragmented risk-factor studies hit a ceiling, and interpretable ML is the way past it. Reading what drives TB outcomes means combining several dimensions of patient data, not inspecting them one by one. The study builds a multi-layer framework over demographic, clinical, diagnostic, and treatment-related variables. Descriptive analysis lays out patient profiles and treatment patterns; supervised models estimate outcome risk; SHAP explanations show which factors moved each estimate. Accuracy goes up, and the output stays legible to providers , who can then see who is at higher risk of failure, why the model says so, and where to step in early with closer monitoring and better management.

Contributions

The DOH-CAR TB-DOTS program runs on the National Tuberculosis Program (NTP) protocol. Case finding is passive and begins at the facility: someone with TB symptoms, a cough that will not quit, weight loss, fever, is sent for testing, sputum smear microscopy, the GeneXpert MTB/RIF test, and a chest X-ray . After diagnosis the patient goes onto the Treatment Card and Master List, gets a registration group, and is paired with a health worker or trained treatment partner who watches each dose over the six- to twelve-month course . Every month, a visit records sputum results, vital signs, and weight, and checks that the patient is keeping up. The outcome is later filed under NTP criteria, Cured, Treatment Completed, Failed, Died, or Lost to Follow-up . Most of this is paper, and risk is read in hindsight. A worsening condition, missed doses, a complication from another illness, these tend to come to light only after they have already bent the course of treatment. By then the high-risk patient was never flagged in time, and the cost is avoidable failures, more deaths, and continued spread of drug-resistant (MDR) TB across the region .

The research presents six major contributions. The first contribution is developing and validating an Integrated Predictive Model utilizing machine learning models that combine training

sets based upon demographic, clinical, diagnostic and treatment related variables to forecast therapy success or failure of PTB patients in CAR. Therefore giving clinicians the opportunity to identify potential problem patients earlier.

Secondly, a reproducible Data Preprocessing & Harmonization Pipeline which takes raw patient information from TB-DOTS clinics throughout the CAR and transforms the datasets for machine learning algorithms' use, using a variety of techniques such as schema normalization; pseudonymous identification of individual patients; automatic correction of errors; and a diagnostic-first approach to handle missing values.

Thirdly, a Descriptive/Exploratory Analysis that describes the study population; provides insight into how the various attributes relate to one another; and identifies the most important predictors of treatment outcomes.

Fourthly, Temporal Modeling to examine time dependent relationships in the monthly monitoring data using Recurrent Neural Networks (RNN), Long Short Term Memory (LSTM) architectures and therefore capture variations that occur over time during treatment. Therefore, temporal modeling is expected to generate predictive accuracy that improves upon prior studies which did not account for temporal changes in monitoring data.

Fifthly, Shapley Additive Explanation (SHAP) were used to measure the degree to which each patient-specific factor contributed to a prediction. SHAP are also useful because they allow clinicians to see what specific factors led to each predicted risk's score and thus help evaluate whether or not a particular recommendation should be implemented for each patient.

Lastly, the validated model was embedded into a Clinical Decision Support system (CDSS) that displays a patient's current level of risk; lists all the factors that contribute to each risk assessment via SHAP; and updates a patient's risk estimate when new follow-up data becomes available.

Research Objectives

This study aims to develop and validate machine learning models that predict treatment success or failure among TB patients in the CAR, Philippines. Combining demographic, clinical, and laboratory variables supports early identification of patients at risk of unsuccessful treatment and provides evidence-based tools for improving TB program outcomes. The broader goal is to enable personalized clinical interventions and efficient allocation of healthcare resources. The research aims to achieve the following objectives:

1. To document and pre-process patient records of PTB cases from TB-DOTS facilities in the CAR through data cleaning, feature engineering, and reporting, so that the dataset is ready for ML applications.
2. To describe the demographic, clinical, diagnostic, and treatment characteristics of PTB patients in the dataset and provide an overview of the study population.
3. To identify variables that influence treatment outcomes over the course of therapy, so that the most important factors affecting patient recovery can be recognized and used to guide treatment adjustments.
4. To add temporal modeling that analyzes time-dependent patterns in patient data, so that changes and trends throughout treatment can be detected, improving the accuracy of outcome predictions and supporting timely clinical interventions.
5. To train and validate supervised learning models for predicting treatment success or failure among PTB patients, so that individuals at risk of poor outcomes can be identified early, while the model maintains high predictive accuracy and reliability when applied to incoming TB cases.
6. To develop a decision support tool that visualizes patient risk levels and predictive outputs.

Scope and Limitations

The dataset includes PTB patients from 2015 to 2025 in the CAR, specifically Baguio City, and excludes those still undergoing treatment at the time of data extraction. Only patients with a defined treatment outcome of either success or failure are included. The data covers demographic information, laboratory tests, diagnostic data, clinical data, treatment regimen, follow-up examinations, drug administration data, and treatment outcomes. The dataset's limits are mostly about sufficiency and quality. Because it comes from existing records, completeness and consistency vary, and some clinical parameters are simply incomplete, which narrows the set of risk factors available and weakens parts of the analysis. It also will not fully mirror the current patient population, since anyone still in treatment was left out. Extraction from DOH-CAR brings its own friction. Across the ten years from 2015 to 2025, record-keeping shifts as the health system and its documentation standards change, so consistency is not guaranteed. And where indicators are self-reported, bias and inconsistency can seep in and dull the model's predictions.

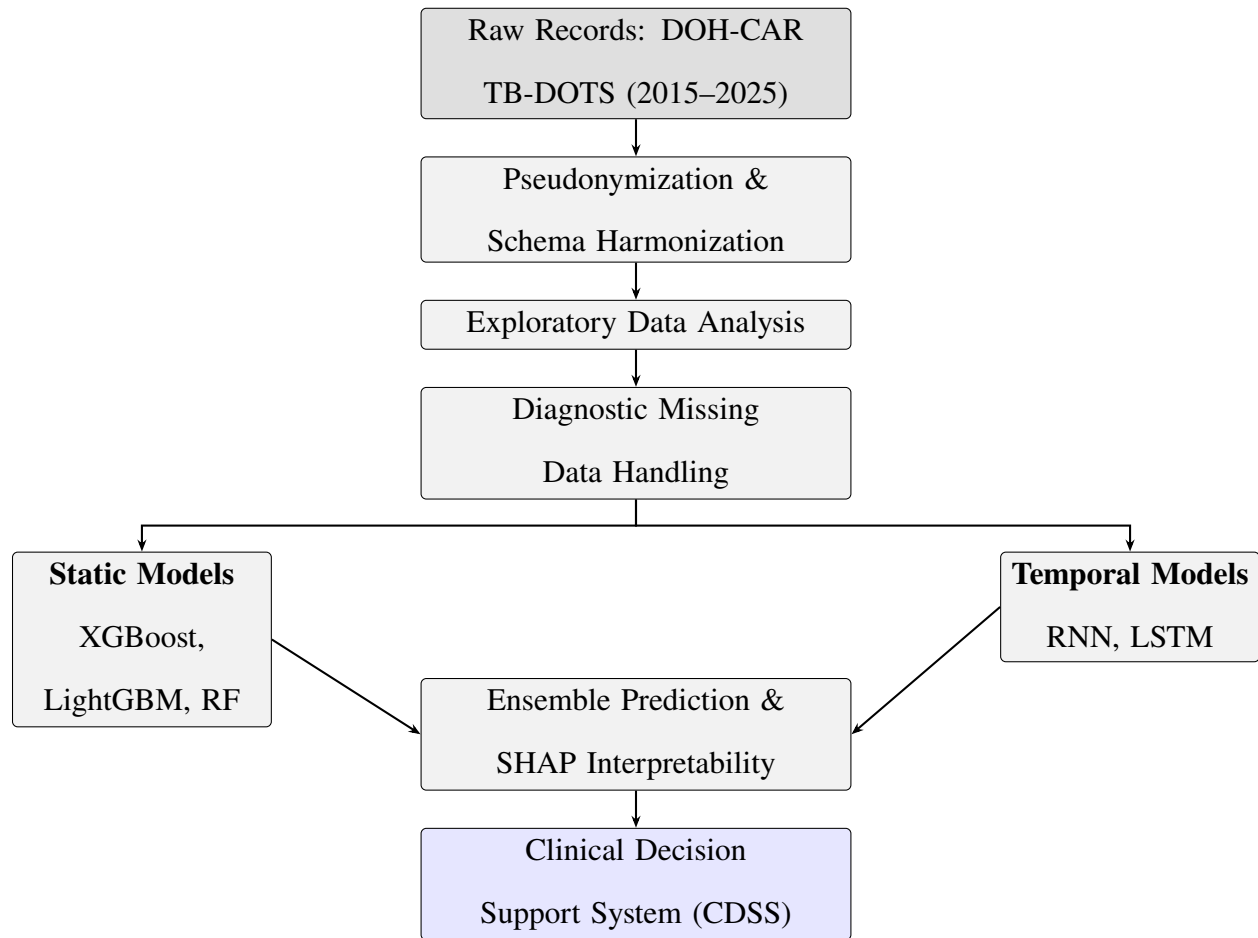
Significance of the Study

The models are meant to help providers spot the patients most likely to fail treatment, early enough for a closer look, tighter monitoring, and better management. Whether clinicians adopt the tool turns on whether they can read and judge its output, so the framework leans on explainable prediction rather than a bare risk score. SHAP surfaces the patient-level factors behind each estimate, and the CDSS lays them out in a form a clinician can actually work from. The larger aim is to strengthen TB control in the Cordillera Administrative Region (CAR) by turning data the program already collects into tools that predict, explain, and can be acted on. Concretely, the results should sharpen patient management and treatment planning, firm up how the program runs, and give public health decisions better footing for lifting the TB treatment success rate in CAR. The study also means to put something usable in administrators' hands, a model that could grow into a working application for the local TB-DOTS program. Alongside it, a cleaned and

pseudonymized derivative of the DOH-CAR PTB Treatment Outcome Dataset will be released for clinical, epidemiological, and machine learning research.

Method

This study draws on PTB patient records from the Cordillera Administrative Region (CAR), Philippines, covering 2015 to 2025 and held centrally by the Department of Health-CAR (DOH-CAR). Work began once ethical approval came through. Records were pseudonymized first, then run through a standardized preprocessing pipeline (schema harmonization, error correction, feature scaling) and the diagnostic-first missing data protocol set out later in the Missing Data Handling subsection. Exploratory and cluster analyses describe the cohort and produce subgroup-level features. Ensemble tree-based classifiers (XGBoost, LightGBM, Random Forest) then learn to predict treatment outcomes as defined by National Tuberculosis Program criteria. Treatment plays out over months rather than a single visit, so temporal models, Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM), are tested beside the static ones and ensembled for robustness. Standard classification metrics measure performance. Shapley Additive Explanations (SHAP) carry interpretability. The final models feed a clinical decision support framework that revises each risk estimate as treatment goes on. Figure 1 gives the overview.

Figure 1*Data Management and Analysis Process***Data Acquisition and Preprocessing**

Two CAR datasets feed the study, both running 2015 to 2025. One is a non-temporal (static) consolidated registry that DOH-CAR maintains; it trains the static tree-based classifiers. The other is a temporal monitoring dataset, compiled by hand by the study’s Pharmacy Co-authors, that trains the longitudinal deep learning and temporal tree-based models. Table 1 puts the two side by side.

Table 1*Comparison of the Non-Temporal and Temporal Monitoring Datasets*

Attribute	Non-Temporal Dataset	Temporal Monitoring Dataset
Records (patients)	7,389	599
Coverage period	2015–2025	2015–2025 (patient subset)
Data source	DOH-CAR centralised TB-DOTS registry	Facility treatment monitoring forms (35 raw Excel workbooks)
Collection method	Routine national TB-DOTS programme reporting	Manual transcription and encoding by Pharmacy Co-authors
Structure	Cross-sectional: one row per patient	Longitudinal: one row per patient-month (M0–M12)
Static features	Demographics, diagnostics, outcomes	Demographics, diagnostics, baseline vitals, outcomes
Temporal features	None	Monthly adherence, doses taken/missed, weight, height, smear microscopy, Xpert MTB/RIF (M0–M12)
Post-preprocessing columns	28	269
Post-preprocessing rows retained	7,388	599 (zero deleted)
Primary model use	Static tree-based classifiers (XGBoost, LightGBM, Random Forest)	Temporal deep learning models (Bi-LSTM) and tree-based temporal baselines

Non-Temporal Dataset

DOH-CAR builds this dataset centrally, pulling reports from government TB-DOTS facilities and from private practitioners in the Public-Private Mix DOTS (PPMD) scheme . There are 7,389 patient records, baseline features only: demographics (age, sex, civil status), clinical indicators (BMI, blood pressure), diagnostic results (GeneXpert, Smear Microscopy). Each row is one patient at the point of treatment registration, with no follow-up monitoring attached. This is the main training source for the static classifiers (XGBoost, LightGBM, Random Forest).

The non-temporal pipeline harmonizes variable formats and units, then pseudonymizes the records. The binary target comes straight from standardized National Tuberculosis Program (NTP) criteria: ‘Cured’ or ‘Treatment Completed’ map to ‘Success’; ‘Failed’, ‘Died’, or ‘Lost to Follow-

up’ map to ‘Failure’. Continuous clinical variables get their extreme outliers capped so training is not skewed. Whatever is missing passes through the diagnostic-first protocol described later in the Missing Data Handling subsection.

Splitting into training and testing sets happens before any transformation, which keeps test information out of the model. The split is stratified, so success and failure proportions stay level across both partitions. Everything downstream, feature scaling, normalization, the multivariate imputation of missing values, is fit on the training set and nothing else. Those fitted parameters and imputation rules then transfer to the held-out test set. No model ever sees the test data’s distribution.

Temporal Monitoring Dataset

The temporal monitoring dataset is a longitudinal subset of 599 patients, each with complete month-by-month records. The DOH-CAR registry held nothing like it. Instead, the study’s Pharmacy Co-authors *gathered and encoded it by hand* from paper Treatment Cards and Master Lists at TB-DOTS centers in Baguio City, Benguet, going to the facilities in person and transcribing each patient’s monitoring data: demographics, monthly vital signs, percentage adherence, doses taken and missed, smear microscopy (TB-LAMP) and Xpert MTB/RIF results, and treatment outcomes. Everything landed in a master RAW DATA.xlsx workbook, one sheet per barangay or facility unit, across 35 facility-year cycles from 2015 to 2025. A consolidation script, `combine_raw_dataset.py`, then merged the per-barangay sheets, normalizing facility names against a canonical alias map so mismatched entries lined up before the main pipeline ran.

The layout is wide: one row per patient, columns holding monthly observations from treatment Month 0 (M0) through Month 12 (M12). Per monthly visit, the columns record percentage adherence (`pct_adherence`), doses taken and missed that month, cumulative doses taken, body weight, height, smear microscopy (TB-LAMP), and the Xpert MTB/RIF result. Static baseline features sit alongside the monthly ones on the same row: demographics, baseline vital signs (heart rate, blood pressure, oxygen saturation, temperature, respiratory rate), comorbidities, the starting

treatment regimen, and diagnostic classification. After preprocessing, the cleaned table is 599 rows by 269 columns.

The temporal pipeline has eight stages, shaped around this wide longitudinal layout:

1. **Schema Validation & Column Standardization.** Column names are forced to snake_case. Duplicates and fully null columns get flagged and dropped.
2. **Identifier Removal.** Four patient- and facility-identifying columns (no, source_file, name_of_diagnosing_facility, name_of_treatment_unit) stay in the export file for traceability but never reach any model input, which is how patient privacy is enforced.
3. **Data Cleaning & Type Coercion.** Every column is parsed to its correct type and clinically validated. Domain-specific formats get their own parsers: weight strings (45kg, 56.5 kg) become numeric kilograms; heights become centimeters; blood pressure in 120/80 form splits into separate systolic and diastolic fields; oxygen saturation arrives as a decimal fraction (0.95), a percent-suffixed value (95%), or a raw integer, and all three collapse to one 0 to 100 scale; adherence percentages lose any trailing %. Dates get a correction pass of their own. A future date of birth is pulled back (e.g., 2063 → 1963), transposed digits are caught (e.g., 2990 → 1990), and ordering is enforced (treatment start ≥ diagnosis date). Continuous clinical variables are then winsorized to physiologically plausible bounds: temperature to 30 to 45°C, systolic BP to 60 to 220 mmHg, diastolic BP to 30 to 140 mmHg, heart rate to 20 to 250 bpm, oxygen saturation to 70 to 100%, weight to 10 to 180 kg, height to 80 to 220 cm, respiratory rate to 4 to 60 breaths/min.
4. **Categorical Harmonization.** Inconsistent categorical entries are mapped onto canonical labels. Outcome variants, sex designations, regimen types, comorbidity flags, civil status: each collapses from hundreds of free-text variants down to its small expected set.
5. **Drop Uninformative Columns.** Columns that are near-entirely null or redundant (e.g., nationality_raw once it has merged into nationality) are flagged in the output, not silently deleted, so the cleaning decisions stay auditable.
6. **Wide → Long Temporal Restructuring.** The wide table is melted into two: a static

baseline table (`df_static`, one row per patient) and a temporal table (`df_temporal`, one row per patient-month) with columns [`patient_id`, `month`, `weight`, `height`, `pct_adherence`, `monthly_doses_taken`, `monthly_missed_doses`, `cumulative_doses_taken`, `smear_tb_lamp`, `xpert_mtb_rif`].

7. **Diagnostic-First Missing Data Handling.** The four-pathway diagnostic pipeline from the Missing Data Handling subsection runs independently on the static and temporal feature sets. Temporal features are never hard-excluded for being sparse. Their missingness is structural, later months simply carry fewer observations as patients finish or abandon treatment, and that pattern is preserved with missingness indicators.
8. **Post-Imputation Clinical Clipping & Export.** Clinical bounds are re-applied to every imputed value (e.g., `weight` to 10 to 180 kg, `pct_adherence` to 0 to 100). A monotonicity repair pass undoes any MICE-introduced dip in `cumulative_doses_taken` within a patient. The pipeline then writes `cleaned_human_readable.csv` (599 rows $\times$ 269 columns), original clinical units and category labels intact. Encoding, scaling, and tensor preparation wait for the model-specific scripts that run after the train-test split, deliberately, so nothing leaks.

Same as the non-temporal case, the temporal train-test split comes before any scaling, one-hot encoding, outlier capping, or 3-dimensional tensor construction for the recurrent networks. Both pipelines guard against leakage in the same way, static classifier or temporal network alike. Table 2 reports per-patient observed means of the temporal features across M0 to M7.

Table 2*Per-Patient Observed Means of Temporal Features (M0–M7)*

Feature	N	Mean	SD	Min	Max
monthly_doses_taken	573	23.36	4.32	0.0	64.29
cumulative_doses_taken	535	70.09	23.39	0.0	127.38
monthly_missed_doses	458	0.91	2.47	0.0	25.0
pct_adherence	342	98.23	8.35	0.0	100.0
weight	361	55.69	21.55	7.2	189.69
height	315	155.49	20.45	31.0	181.0
smear_tb_lamp	25	0.12	0.33	0.0	1.0
xpert_mtb_rif	41	0.0	0.0	0.0	0.0
last_month_attended	599	5.23	2.06	-1.0	7.0
total_observed_months	599	6.21	2.06	0.0	8.0

Exploratory Data Analysis

An exploratory pass came before any modeling. It profiled the study population, gauged how complete the data was, and measured how the variables associated with each other. Those findings shaped the preprocessing and modeling choices that followed.

Table 3*Missing Data Summary by Column*

Column	Missing_Count	Missing_Percentage
Days_To_Microscopy_Result	7297	98.75
Days_To_RDT_Result	6638	89.84
Days_Screening_To_Diagnosis	4444	60.14
Days_To_Treatment	4378	59.25
Diagnosis_Season	3953	53.5
Diagnosis_DayOfYear	3953	53.5
Diagnosis_Quarter	3953	53.5
Diagnosis_Month	3953	53.5

Column	Missing_Count	Missing_Percentage
Microscopy Result	2792	37.79
Brgy	1026	13.89
City/Municipality	105	1.42
Province	71	0.96
Treatment Health Facility	29	0.39
Region	15	0.2
Date of Outcome/Status	2	0.03
Bacteriologic Status	1	0.01
Type	0	0
Validation Status	0	0
Age	0	0
Sex	0	0
Anatomical Site	0	0
Source of Patient	0	0
Screening/Diagnosing Health Facility	0	0
RDT Result	0	0
Year	0	0
Outcome/Status	0	0
Registration Group	0	0
Computed_Age	0	0

A global missingness audit ran over all 28 columns and ranked each by its missing rate (Table 3). Some variables were badly incomplete. Microscopy result was missing 37.8% of the time, the diagnosis-derived temporal features 53.6%, and the clinical delay intervals anywhere from 59.3% up to 98.8%. Those numbers fixed the missingness thresholds applied later in the Missing Data Handling subsection.

For the continuous variables, pairwise Pearson's r was taken on available-case observations: age, four derived clinical delay intervals (days from screening to diagnosis, to treatment, to microscopy result,

and to RDT result), and diagnosis day-of-year. The question behind it was simple, whether MICE was worth the trouble. When continuous predictors actually share correlation structure, MICE can borrow signal from the observed features to estimate the missing ones. When they don't, a plain univariate mean-fill does the same job.

Categorical variables used Cramér's V instead, a chi-square-based measure scaled to the $[0, 1]$ interval. Any column with fewer than two unique non-null values, or fewer than ten non-null observations, dropped out of the matrix. The set here was patient demographics (sex, type of TB, anatomical site) and clinical classifications (registration group, bacteriologic status, microscopy results).

The exploratory phase also clusters patients across demographic, clinical, and diagnostic dimensions to surface latent subgroups. K-prototypes does the work, since it handles the dataset's mix of categorical and continuous variables. The number of clusters is settled iteratively from the Silhouette score and the elbow method together. That lets the analysis separate groups by comorbidity profile, adherence behavior, and how they respond to a regimen.

The cluster assignments do double duty: descriptive interpretation, and engineered features for the supervised models. The labels are one-hot encoded as extra categorical inputs for the tree-based classifiers (XGBoost, LightGBM, Random Forest) and folded into the feature space for the temporal networks (RNN, LSTM). With them, a model can latch onto subgroup-level patterns behind treatment success or failure, not just individual predictors.

The temporal monitoring dataset got the same scrutiny before preprocessing, a global missingness audit over every feature, columns ranked by missing rate. Above 80% missingness, a column was excluded outright. Between 50% and 80%, it went to the missing-indicator-with-fill pathway, weighed against its predictive importance. Threshold triage like this is standard. It fits the handling strategy to how bad the missingness is, rather than forcing one rule across the board.

Patient retention was traced as the share of patients with non-missing adherence data at each interval from M0 through M12. A raw monthly count tells you little, because retention falls naturally over a six- to twelve-month course, so the decay was expressed as a percentage of the starting cohort at each successive month. This also sets the temporal modeling scope. Once fewer than 25% of patients still have data, that month and every month after it drops out of the feature-level correlation analyses, which keeps a small, non-representative surviving group from driving the estimates.

The same Pearson's r check, again on available-case observations, ran over the vital signs in the temporal dataset. A multivariate imputation only earns its keep when the vitals correlate strongly. If they move together, Multiple Imputation by Chained Equations (MICE) can borrow signal from the observed features to fill gaps; if they don't, MICE buys nothing over mean- or median-fill. The vitals here were age, heart rate, oxygen saturation, systolic and diastolic blood pressure, weight, height, respiratory rate, and temperature, wherever the records carried them.

To see how the temporal features relate, Pearson correlations were taken on per-patient observed means across Months 0 through 7. Each feature, a dose metric (monthly doses taken, cumulative doses taken, percentage adherence), an anthropometric one (weight, height), or a diagnostic result (smear microscopy, Xpert MTB/RIF), was averaged across a patient's available visits. Retention characteristics came from the same window: the last month attended and the total months observed. Averaging this way is deliberate, because it preserves the dropout signal. A patient who quits after M4 contributes means from M0 to M4 and no further, and that short window carries information of its own rather than a gap waiting to be imputed.

Categorical variables in the temporal dataset again went through Cramér's V , the chi-square-based measure on the $[0, 1]$ interval. Pearson and Cramér's V were never mixed into a single matrix. Pearson's r assumes interval-level measurement and gives misleading values on nominal or ordinal categories whose spacing is uneven, so keeping Cramér's V separate for the categorical features avoids that and yields interpretable estimates. Columns with fewer than two unique non-null values, or fewer than ten non-null observations, were left out; association can't be estimated meaningfully there anyway.

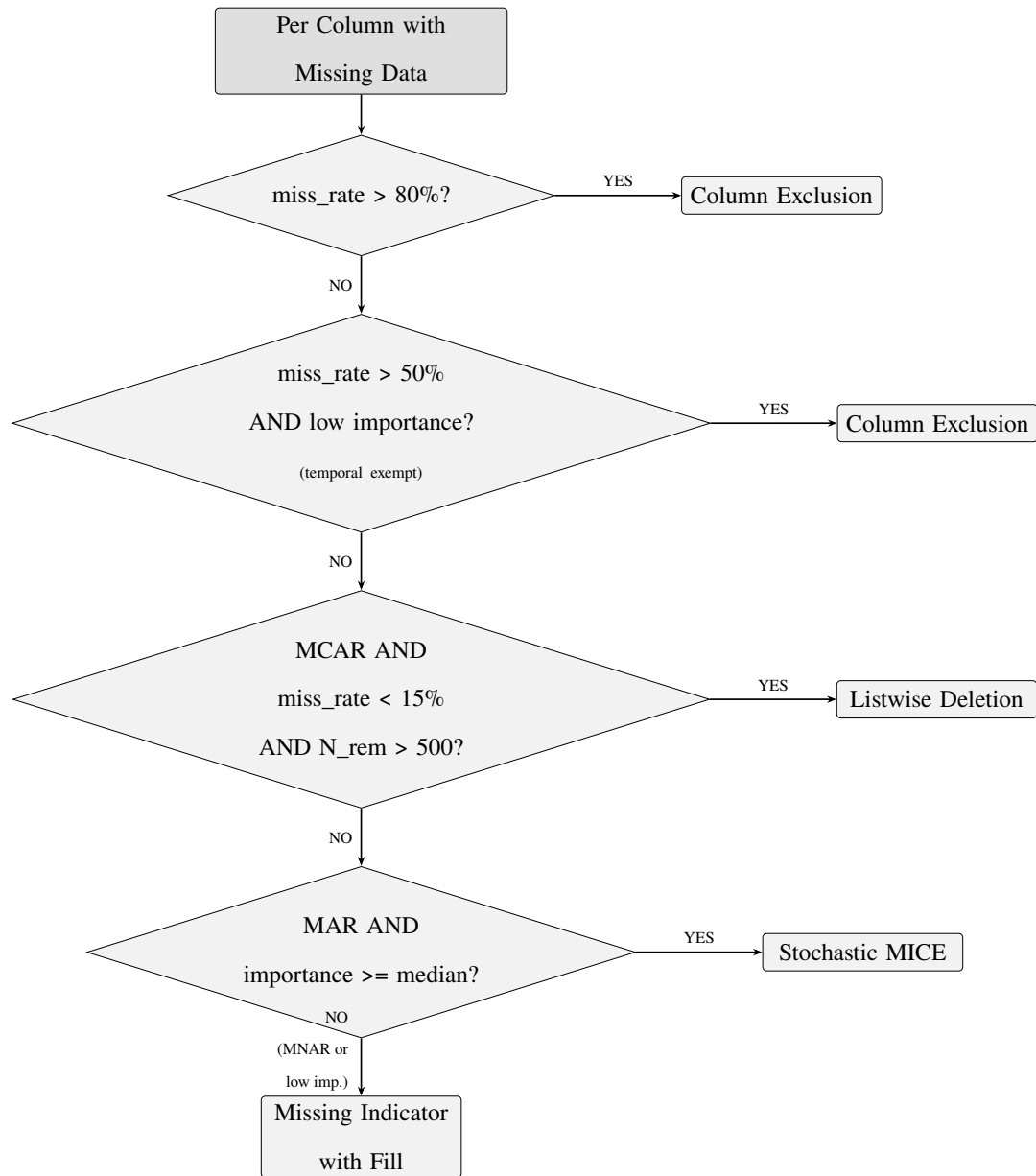
A cardinality audit over every categorical feature in the raw dataset showed how much inconsistent coding had to be fixed. Counting the unique raw values per feature put a number on the entry inconsistency. Where a feature's cardinality ran far past its expected count of valid categories, say a clinical observation field that should carry a handful of standardized codes but instead carried hundreds of free-text variants, it was flagged for harmonization in Stage 4. The audit hands those harmonization targets an empirical baseline: it measures the data entry problem before any cleaning touches it.

Whether adherence holds steady over the course was tested with lag-1 Pearson correlations of percentage adherence across consecutive month pairs (M0 to M1 through M7 to M8). That month-to-month strength decides whether a recurrent architecture, which assumes each time step leans on the one before, is even worth building. Weak correlation between adjacent months would mean a static model on aggregated

patient means tells you just as much. Only strong correlation justifies the extra complexity of an RNN or LSTM .

Missing Data Handling

Missingness in clinical TB registries is uneven. It traces to inconsistent documentation, to facilities of differing capacity, and to the simple fact that treatment is monitored over many months. Once the mechanism behind the gaps shifts from feature to feature, imposing a single imputation strategy on all of them is statistically wrong. So this study runs a diagnostic-first pipeline: it classifies how each variable went missing, then sends that variable down one of four routes, column exclusion, listwise deletion, a missing indicator with fill, or stochastic Multiple Imputation by Chained Equations (MICE) (Figure 2) .

Figure 2*Missing Data Handling Decision Logic*

The first diagnostic step is Little's MCAR test at the dataset level, asking whether the data are Missing Completely at Random (MCAR). Were that true, missingness would be independent of every value, observed or not, the complete cases would be an unbiased subsample, and listwise deletion would be safe. It is not true here. The test came back significant ($p < 0.05$), which rejects MCAR and ties the missingness systematically to the observed data. Deleting cases wholesale is off the table .

Little's test returns one omnibus verdict for the whole dataset, so each column's mechanism still needs its own classification. The pipeline checks every variable's missingness indicator against the observed data. For temporal (monthly) variables, Spearman rank correlations relate the column's monthly missing rate to treatment month; a correlation that climbs monotonically signals a dropout pattern, which is Missing Not at Random (MNAR). For static baseline variables, point-biserial correlations relate the binary is-missing indicator to the observed features. A significant association means other observed variables explain the gap, making it Missing at Random (MAR); a non-significant one points back to MNAR. Classifying column by column, rather than swallowing one dataset-level verdict, lets each handling strategy match the statistics of its own variable .

Two decisions, column exclusion and MICE eligibility, lean on a Random Forest classifier trained on available cases, the rows where the treatment outcome is observed. For ranking alone, features are filled with column medians. It is a coarse fill, but good enough to estimate importance. Variables in the bottom quartile of importance become candidates for soft exclusion; those above the median get priority for stochastic imputation .

A hard threshold drops any variable over 80% missing on sight, importance aside, because nothing reliable can be imputed at that sparsity. A soft threshold also drops variables past 50% missing whose importance falls in the bottom quartile, since imputing a sparse low-value variable adds more synthetic variance than information. Temporal variables sidestep both thresholds. Their structural missingness is the longitudinal design of TB monitoring showing through, not a data quality fault .

Listwise deletion stays on the table for one narrow case: a variable confirmed MCAR at the column level, under 15% missing, with more than 500 patients still left after deletion. In that situation the complete cases are an unbiased subsample, so deletion gives unbiased estimates without the variance distortion that imputation introduces. No variable here cleared all three conditions at once, so the pathway never fired .

Variables classified MNAR, along with every temporal feature regardless of mechanism, get a two-part treatment. First, a binary indicator (`is_missing_[variable]`) marks the absence itself. Then the original variable is filled, last observed value carried forward for temporal variables, column median for continuous static ones, column mode for categorical static ones. The absence is treated as signal, not noise to be papered over. A missing Month 9 weight in TB monitoring usually means the patient dropped out, and the indicator passes exactly that clinical pattern to the model. Backward-fill is barred from temporal

imputation: letting a future measurement seep into an earlier time step would leak information the model has no legitimate access to at inference .

Four vital signs cleared both bars, confirmed MAR and ranked above median importance: oxygen saturation, heart rate, and systolic and diastolic blood pressure. Each is imputed with stochastic MICE under an ExtraTreesRegressor as the conditional model. ExtraTreesRegressor is a nonparametric tree-based estimator. It captures non-linear relationships in mixed-type clinical data without standardized variables, much like the missForest framework , and on this kind of data it usually outperforms linear and k -nearest-neighbor imputers. The MICE iteration count tracks each variable's missing fraction, scaling as $\max(20, \min(100, \lfloor \text{miss_rate} \times 100 \rfloor))$, which gives high-sparsity features room to converge without letting compute run away. That mirrors Von Hippel's rule that the number of imputations should roughly match the percentage of missing information , applied here to iterations instead of to the count of imputed datasets. And imputing stochastically rather than deterministically keeps the natural spread of the imputed variables, instead of collapsing them onto conditional means .

Data Quality Assurance

A layered quality assurance protocol keeps the supervised pipeline reliable. It runs across schema validation, clinical bounds checking, temporal consistency, and post-processing validation.

Schema and Column Validation

Column names are normalized to snake_case: whitespace stripped, text lowercased, special characters swapped for word equivalents, so `M0 %Adherence` turns into `m0_pct_adherence`. Duplicate columns and fully null ones are flagged and deduplicated before modeling. Four patient- and facility-identifying columns remain in the export file for traceability, but none of them reach a model input, for privacy.

Date Integrity Correction

A dedicated correction pass deals with the recording errors that clinical registries tend to accumulate. A future date of birth, its year past the data year, is pulled back 100 years. A date of birth before 1920 is pushed forward 100 years when that puts the age in a plausible 0 to 110 range. Transposed digits, 2990 for 1990 and the like, are caught against adjacent records. Date pairs are forced into order, treatment start no earlier than diagnosis, outcome date no earlier than treatment start. A missing date inherits the cohort median date for its variable.

Temporal Consistency

Every fill in the longitudinal structure is forward-only. That alone keeps the temporal data from leaking. A Month 4 observation can carry forward to Month 5; a Month 6 measurement can never inform an earlier step the model is meant to predict without it. After MICE, a monotonicity repair pass reverses any drop in `cumulative_doses_taken` that the multivariate imputer introduced, so a patient's cumulative dose never falls from one treatment month to the next.

Categorical Harmonization Audit

A cardinality audit of the raw categorical features confirmed how widespread the entry inconsistency was. The `chest_x_ray_at_case_notification` field carried over 300 unique free-text entries, `co_morbidities` more than 150; even nominally binary fields like `sex` and `outcome` showed 5 to 12 raw representations each, the residue of treatment centers abbreviating in their own way. Those counts drove a 17-field harmonization pass that maps free-text and inconsistently coded entries onto standardized categories before analysis.

Missing Data Diagnostics and Feature Importance

The diagnostic-first missing data pipeline, detailed in the Missing Data Handling subsection, adds one more validation layer. Little’s MCAR test runs at the dataset level. Per-column mechanisms come from two correlation checks, Spearman rank to catch temporal dropout, point-biserial to separate MAR from MNAR. A Random Forest trained on available cases ranks predictive importance, and that ranking decides where each variable is routed.

Post-Processing Verification

Before the cleaned dataset reaches any model, an automated routine checks four things: (1) the identifier columns are still present for traceability, (2) no scaled or normalized values have slipped in, so everything stays in original clinical units, (3) categorical labels are still human-readable strings, not integer codes, and (4) no one-hot columns have appeared. Pass all four, and encoding, scaling, and tensor construction were demonstrably left to the model-specific scripts that run after the split.

Leakage Prevention and Reproducibility

Every validation step is there for one reason: keep test-set information out of training. The patient-level stratified split comes first, before any transformation. Scalers, imputers, and encoders are fit on the training set alone, then carried to the test and validation sets through `reindex()` so the column sets line up exactly. Temporal fills run forward-only, backward-fill switched off. Reproducibility is handled two ways: the split is fully deterministic under a seeded random state, and the fitted scaler parameters are checked by SHA-256 hash to confirm that independent runs preprocess identically. An audit trail, `missing_data_report.json`, logs the decision pathway, mechanism classification, and rationale for every column.

Identification of Variables Associated With Treatment Outcomes

To find which variables track with treatment outcomes, the study pairs descriptive association analysis with model-derived explanation. The descriptive side summarizes which demographic, clinical, diagnostic, and temporal variables pull the success and failure groups apart. The supervised models then estimate how those same variables combine to predict the outcome. Both are needed, because risk in TB treatment is multidimensional. A variable can look weak on its own and still matter once it sits next to adherence, diagnostic, geographic, or monitoring information.

Connecting predictive performance to clinical interpretability falls to Shapley Additive Explanations (SHAP), a post-hoc method that measures each variable’s marginal contribution to the model’s output. SHAP shows which features pushed a patient-level prediction toward “Success” (Cured, Completed) or “Failure” (Failed, Died, Lost to Follow-up;). In clinical review, that amounts to seeing what the model leaned on, drug resistance, comorbidities, adherence gaps, diagnostic results, a delayed diagnosis. One caution holds throughout: these explanations describe how the model behaved, not clinical causality, and the line between computational attribution and medical judgment is kept deliberately clear.

Temporal Modeling

TB treatment runs 6 to 12 months, so the study adds temporal modeling on the longitudinal monitoring data. Train a static model on baseline features alone and it can miss how adherence, follow-up attendance, and lab measurements move over the course. The temporal models are built to give month-by-month risk estimates from treatment Month 0 (M0) through Month 12 (M12).

These models run on one dataset only, the 599-patient longitudinal set from the Data Acquisition and Preprocessing subsection (see also Table 1), the one the Pharmacy Co-authors compiled by hand from facility monitoring forms. It is not the 7,389-patient DOH-CAR registry the static classifiers use; the two are separate. After the eight-stage temporal pipeline, each patient has up to 8 usable monthly timesteps, M0 to M7. Months past M7 are dropped from the model inputs. Fewer than 25% of patients had observations that far out (Table 5), and a surviving subset that small and skewed gives unreliable estimates. As before, encoding, scaling, and 3-dimensional tensor preparation wait for the post-split, model-specific scripts, so

nothing leaks.

Temporal modeling went through two iterations, v1 and v2. They share the same dataset lineage; v2 is the corrected, modeling-ready preparation that came out of a methodological review of v1. The v1 pipeline was a modeling baseline, with patient-level stratified splitting and progressive temporal expansion across M0 to M12. The review then flagged a few things that could weaken its clinical defensibility: late-treatment fields that raise the risk of outcome leakage, inconsistent handling of imputed or unreliable outcome labels, and cleaning and temporal feature construction that had not been audited closely enough.

Version 2 (v2) fixed those issues and retrained. It builds on the audited output of the eight-stage cleaning pipeline from Data Acquisition and Preprocessing, restricts supervised learning to patients whose outcome is actually observed, and removes leakage-prone and administrative fields, outcome-date fields and late-timeline variables that would not exist at an early inference. Temporal predictors are encoded two ways: as sequence inputs for the deep models, as engineered summaries for the tree-based ones. The deep architecture is a Hybrid Bidirectional LSTM (Bi-LSTM), picked to model long-term dependencies without the vanishing gradients of a plain RNN ; its recurrent structure suits the strong month-to-month adherence dependence the lag-1 analysis turned up . The tree-based baselines take the other route, flattening the monthly values and bolting on aggregates, trends, latest-observed values, and missingness indicators. Class imbalance is corrected with training-only resampling such as SMOTE-ENN. The hybrid framing is kept so temporal predictions can sit beside strong static learners like XGBoost, which holds up better across the different patient subgroups .

Supervised Learning Model Development and Validation

At the center of the study, supervised models are trained and validated to classify treatment outcomes. The algorithms are ensemble tree-based ones, which handle complex non-linear interactions among mixed-type clinical variables and take class imbalance in stride: XGBoost , LightGBM , and Random Forest . The three came out of a preliminary run across six candidates that also included Logistic Regression, SVM, and Gradient Boosting. The three that lost out either discriminated poorly ($CV-AUC \leq 0.63$) or recalled almost none of the minority failure class, either of which makes a model unsafe inside a CDSS. Each surviving classifier runs under two sampling strategies: no resampling with native class-weighting, and

SMOTE with Edited Nearest Neighbors (SMOTE-ENN), which inflates the training failure class by 433% to a near-balanced 0.70:1 ratio.

Feature Set Versions

Three feature sets are evaluated in sequence, to see how much each extra clinical and temporal block actually buys. The *baseline* set (11 features) is the core demographic and diagnostic block: age, sex, anatomical site, bacteriologic status, microscopy result, registration group, source of patient, TB type, province, days to treatment, and year of registration. The *improved* set (14 features) adds three facility-level geographic variables: city/municipality, treatment health facility, and screening/diagnosing health facility. The *extended* set (21 features) adds a further seven temporal and diagnostic delay features from date engineering: diagnosis month, diagnosis quarter, diagnosis season (on the Philippine three-season meteorological calendar: *Tag-lamig* for cool dry December to February, *Tag-init* for hot dry March to May, *Tag-ulan* for the rainy June to November), days from screening to diagnosis, days from diagnosis to microscopy result, days from diagnosis to rapid diagnostic test (RDT) result, and the RDT result itself (binned from 42 raw variants into three classes: Detected, Not Detected, Not Done). Date of outcome was kept out of every feature set. That field is generated after the treatment result, so including it would leak the outcome.

Data Partitioning and Class Imbalance

A stratified 70/20/10 patient-level split divides the data into training (70%), test (20%), and holdout validation (10%), with the class proportions held across all three. The imbalance is heavy, roughly 86% success against 14% failure, and it is corrected on the training set only, with one of three oversamplers. SMOTE interpolates synthetic minority-class points. SMOTE-ENN runs SMOTE, then clears ambiguous borderline cases with Edited Nearest Neighbors undersampling. SMOTE-Tomek runs SMOTE, then removes Tomek links to tidy the boundary. The test and validation sets are never resampled. They keep their original distributions, so the evaluation stays unbiased.

Cross-Validation and Model Selection

Every configuration is scored by five-fold stratified cross-validation on the training data, which estimates generalization and returns a mean and standard deviation of ROC-AUC across folds . The clinical goal is to catch at-risk patients early, so selection runs on a composite metric, $\text{score} = 0.6 \times \text{AUC} + 0.4 \times \text{Recall(Failure)}$: it leans on discrimination but still rewards sensitivity to the minority failure class. Whichever configuration tops that score across the feature versions and resamplers is then checked on the held-out 10% validation set for an unbiased real-world estimate.

Each configuration also reports the Matthews Correlation Coefficient (MCC) next to the composite metric, as a supplementary check. Accuracy or a class-specific F1 can mislead; MCC reads all four quadrants of the confusion matrix at once and holds up under heavy imbalance. It sits in $[-1, +1]$: +1 for perfect prediction, 0 for random, -1 for complete inversion . Putting MCC beside the composite score exposes the trade-off that aggressive resampling brings, since SMOTE-ENN and its kind can lift failure recall while piling on false positives that push overall error up.

Performance Evaluation

Performance is read across several metrics, each one catching a different facet of classification quality under imbalance. Treatment failure is only 13.8% of the cohort, so leaning on a single number, accuracy most of all, can mislead. A classifier that answers “Success” every time scores 86.2% accuracy and catches not one failure. The framework therefore reports discriminative, threshold-dependent, and calibration-aware metrics side by side.

Every metric traces back to the confusion matrix, with treatment failure as the positive class. On the held-out test set, each model produces four quadrants: true positives (TP, failures correctly predicted), false negatives (FN, failures missed), false positives (FP, successes wrongly flagged as failures), and true negatives (TN, successes correctly identified). The metrics below all build on these four:

Accuracy ($\frac{TP+TN}{TP+TN+FP+FN}$) is the overall share of correct predictions. Intuitive, yes, but under 86:14 imbalance the majority class dominates it, so it is reported with that caveat attached.

Precision ($\frac{TP}{TP+FP}$), or positive predictive value, is the share of failure alerts that turn out to be real

failures. When precision is low, false alarms pile up, and clinician trust in the CDSS goes with them.

Recall ($\frac{TP}{TP+FN}$), also sensitivity or the true positive rate, is the share of actual failures the model catches. In TB-DOTS a missed failure costs more than a false alert: an undetected treatment breakdown can run on into disease progression, drug resistance, and further community transmission.

Specificity ($\frac{TN}{TN+FP}$), the true negative rate, is the share of actual successes correctly identified. Keep specificity high and patients who are doing fine are not pulled in for needless review.

F1 Score ($2 \times \frac{Precision \times Recall}{Precision + Recall}$) is the harmonic mean of precision and recall, a single number that punishes a lopsided split between the two. It is handy for comparing models across resampling strategies.

Balanced Accuracy ($\frac{Recall + Specificity}{2}$) averages sensitivity and specificity, which makes it robust to class imbalance by construction. At 0.50 it is random guessing, whatever the class distribution.

Area Under the Receiver Operating Characteristic Curve (ROC-AUC) measures discrimination across every possible decision threshold. The ROC curve plots true positive rate against false positive rate at each threshold; the AUC collapses that curve to one scalar. 1.0 is perfect separation, 0.5 is chance. Because it does not depend on a single threshold, ROC-AUC is the primary ranking metric here, though it can read optimistically under severe imbalance, where the large majority class dominates the false positive rate.

Area Under the Precision-Recall Curve (PR-AUC) answers that limitation, plotting precision against recall across thresholds and ignoring true negatives outright. When the positive class is rare, PR-AUC tells you more than ROC-AUC. Its no-skill baseline is just the positive-class prevalence (0.138), a natural reference point.

Matthews Correlation Coefficient (MCC) is computed as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

MCC takes in all four confusion-matrix quadrants at once and stays informative under class imbalance. The coefficient runs in $[-1, +1]$: +1 perfect, 0 random, -1 fully inverted. Unlike accuracy or F1, it is symmetric under class swapping and does not inflate when the majority class dominates.

Threshold optimization. Under heavy imbalance a 0.50 cutoff need not line up with clinical priorities, so each model gets a threshold search. Candidates from 0.10 to 0.90 are scored against composite objectives, each encoding a different priority: *balanced* (40% F1, 30% recall, 30% specificity), *failure_recall*

(55% failure recall, 25% balanced accuracy, 20% F1), and *specificity* (50% specificity, 30% F1, 20% recall).

A model reports its metrics at whichever threshold maxes out the relevant objective.

Model selection uses a composite score weighting discriminative performance and clinical utility:

$$S = 0.6 \times \text{ROC-AUC} + 0.4 \times \text{Recall(Failure)}$$

Most of the weight sits on ranking ability (AUC), but the score still rewards sensitivity to the minority failure class. Whichever configuration tops it becomes the recommended model.

Validation strategy. Performance is estimated at three levels, to guard against overfitting and to gauge generalization. Five-fold stratified cross-validation on the training set gives the mean and standard deviation of ROC-AUC across folds, a stability check. The 20% held-out test set carries the primary evaluation behind every metric above, on data the model never met in training or preprocessing. A final 10% holdout is reserved for the recommended model alone, to confirm the test-set numbers are not an artifact of one split. Scaling, encoding, and imputation are all fit on the training set, then applied to the test and validation sets, so no evaluation data feeds back into training.

For reproducibility, every experiment goes through one centralized pipeline that standardizes training, cross-validation, and evaluation. Metrics and visualizations export straight to native $\text{\LaTeX}$ for this manuscript.

Development of a Clinical Decision Support Tool

The validated models are embedded in a Clinical Decision Support System (CDSS) built for health-care providers, each part with its own job. The predictive model estimates the probability of treatment success or failure. The SHAP layer names the factors behind that estimate. The interface displays the risk and those factors, and the clinician reads the whole thing against what they already know about the patient. In practice the provider opens a dashboard: the failure and success probabilities, a risk-factor view that surfaces what drove the score (adherence gaps, lab results, diagnostic delays), and live updates. Enter new follow-up data, a monthly weight or a smear result, and the prediction refreshes, so review carries on across the whole treatment. The CDSS is for risk stratification and explanation. It does not diagnose or prescribe

on its own.

Small Language Model (SLM)-Based Narrative Explanation Layer

To make the explanations easier on clinical users, the CDSS adds a small language model (SLM), a constrained narrative layer that renders model outputs as plain clinical text. The SHAP feature contributions remain the explanation of record. The SLM is a presentation layer and nothing more, not a predictor, not a diagnostic engine, not a source of fresh clinical inference . It runs after prediction and touches nothing upstream: not the training, not the inference probabilities, not the risk thresholds, not the SHAP values. In the implemented pipeline, the web client assembles a structured interpretation request, selected patient context fields, the predicted treatment-failure probability, and the ranked contribution signals, then posts it to a dedicated endpoint that streams the narrative back into the interface.

The SLM stays confined to explanation. It does not infer new clinical causes, write prescriptions, suggest regimen changes, or stand in for a clinician’s judgment. Its one job is to make the SHAP-grounded attribution easier to read, so a provider can see why the CDSS flagged a patient and decide whether a closer look is warranted. Keeping the roles this separate matches how medical-domain SLMs are framed today, and how health AI governance expects bounded decision support to behave, and it keeps interpretability tied to the original attribution signal . The final call still belongs to the treating professional.

Results and Discussion

This section presents the results of the machine learning analyses for predicting treatment success and failure among PTB patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes. Predictive model performance is evaluated on the test and validation datasets. Both static tree-based models (XGBoost, LightGBM, Random Forest) under two sampling conditions and temporal models are compared. Feature importance analyses provide evidence about the factors that influenced model predictions. The discussion interprets these findings in the context of clinical relevance, public health priorities, and CDSS usability, and shows how risk estimates and explanation outputs can support early intervention, resource allocation, and stronger TB management.

Missing Data Handling Outcomes

Of the 44 features identified with missing data, 15 were excluded via column exclusion. All 15 exceeded the hard missingness threshold of 80%, including near-complete variables such as tuberculosis culture (99.8%), other laboratory tests (99.8%), and tuberculin skin test (93.5%). One more variable, the raw blood pressure string field, was soft-excluded at 67.3% missingness and below-median predictive importance because it had already been parsed into the structured systolic and diastolic blood pressure variables. All 599 patient records were kept in the dataset, with zero row deletion. Full sample retention matters given the small cohort size and the severe class imbalance of 86.2% treatment success versus 13.8% failure. Listwise deletion under these conditions would have made the minority class representation problem worse .

The listwise deletion pathway was not activated. The dataset-level MCAR test was rejected, and no individual variable met all three conditions of confirmed MCAR mechanism, fewer than 15% missing observations, and a post-deletion sample size above 500. Twenty-five variables were routed to the missing indicator with fill strategy: eight temporal monthly features (including weight, percentage adherence, cumulative doses taken, and monthly smear results) and 17 static baseline variables with confirmed MNAR patterns. Four variables (oxygen saturation, heart rate, systolic blood pressure, and diastolic blood pressure) were imputed via stochastic MICE after confirmation of a MAR mechanism. Their missingness indicators were significantly associated with visit attendance under the point-biserial correlation procedure described above. This is a clinically plausible MAR mechanism, since vital sign recording depends on whether the patient attended the monthly monitoring visit rather than on the unobserved values of the measurements themselves. The concentration of stochastic imputation on this small set of high-value features reflects the diagnostic-first design of the pipeline, which reserves computationally intensive probabilistic imputation for variables where the assumptions required for its validity are met .

Table 4*Global Feature Missingness Rates in the Raw Temporal Dataset*

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m8_weight	599	100.0	Yes
m12_weight	599	100.0	Yes
m11_height	599	100.0	Yes
m9_xpert_mtb_rif	599	100.0	Yes
m9_smear_tb_lamp	599	100.0	Yes
m7_xpert_mtb_rif	599	100.0	Yes
m7_smear_tb_lamp	599	100.0	Yes
m10_smear_tb_lamp	599	100.0	Yes
m10_weight	599	100.0	Yes
risk_factor_s_for_drug_ _resistance_tuberculosis	599	100.0	Yes
m12_height	599	100.0	Yes
m12_smear_tb_lamp	599	100.0	Yes
m12_xpert_mtb_rif	599	100.0	Yes
m9_weight	599	100.0	Yes
m8_xpert_mtb_rif	599	100.0	Yes
m8_smear_tb_lamp	599	100.0	Yes
m1_xpert_mtb_rif	599	100.0	Yes
m11_weight	599	100.0	Yes
m10_xpert_mtb_rif	599	100.0	Yes
m11_smear_tb_lamp	599	100.0	Yes
m11_xpert_mtb_rif	599	100.0	Yes

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m3_xpert_mtb_rif	599	100.0	Yes
m4_xpert_mtb_rif	598	99.83	Yes
m10_height	598	99.83	Yes
tuberculosis_culture	598	99.83	Yes
m12_monthly_missed_doses	598	99.83	Yes
other_lab_test	598	99.83	Yes
m12_pct_adherence	598	99.83	Yes
m8_height	598	99.83	Yes
m9_height	598	99.83	Yes
m12_monthly_doses_taken	597	99.67	Yes
m4_smear_tb_lamp	597	99.67	Yes
m1_smear_tb_lamp	597	99.67	Yes
others	597	99.67	Yes
m11_monthly_missed_doses	597	99.67	Yes
m11_pct_adherence	597	99.67	Yes
m12_cumulative_doses_taken	597	99.67	Yes
m10_monthly_missed_doses	596	99.5	Yes
m9_pct_adherence	596	99.5	Yes
m10_pct_adherence	596	99.5	Yes
m11_monthly_doses_taken	595	99.33	Yes
m9_monthly_missed_doses	595	99.33	Yes
m11_cumulative_doses_taken	595	99.33	Yes
m8_pct_adherence	593	99.0	Yes
m3_smear_tb_lamp	593	99.0	Yes
m8_monthly_missed_doses	592	98.83	Yes
m10_monthly_doses_taken	592	98.83	Yes

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m10_cumulative_doses_taken	592	98.83	Yes
m6_smear_tb_lamp	590	98.5	Yes
m9_cumulative_doses_taken	588	98.16	Yes
m7_height	588	98.16	Yes
m5_smear_tb_lamp	588	98.16	Yes
m9_monthly_doses_taken	588	98.16	Yes
m6_xpert_mtb_rif	587	98.0	Yes
m2_smear_tb_lamp	587	98.0	Yes
m0_smear_tb_lamp	586	97.83	Yes
m7_weight	586	97.83	Yes
m5_xpert_mtb_rif	585	97.66	Yes
m8_cumulative_doses_taken	582	97.16	Yes
m8_monthly_doses_taken	581	96.99	Yes
dat_supported_dup	576	96.16	Yes
m2_xpert_mtb_rif	572	95.49	Yes
other	570	95.16	Yes
m5_height	563	93.99	Yes
m4_height	562	93.82	Yes
m6_height	562	93.82	Yes
m6_weight	561	93.66	Yes
tuberculin_skin_test	559	93.32	Yes
m3_height	558	93.16	Yes
m5_weight	557	92.99	Yes
m2_height	555	92.65	Yes
m0_xpert_mtb_rif	554	92.49	Yes
m4_weight	554	92.49	Yes

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m1_height	554	92.49	Yes
nationality_raw	551	91.99	Yes
m3_weight	550	91.82	Yes
dat_supported	549	91.65	Yes
prior_history_of_tb	549	91.65	Yes
m2_weight	543	90.65	Yes
m1_weight	540	90.15	Yes
m7_pct_adherence	519	86.64	Yes
regimen_type_at_6th_month_of_ _treatment	518	86.48	Yes
smear_microscopy	502	83.81	Yes
m7_monthly_missed_doses	502	83.81	Yes
respiratory_rate	501	83.64	Yes
height	492	82.14	Yes
temperature	481	80.3	Yes
weight	476	79.47	No
m7_cumulative_doses_taken	465	77.63	No
o2_sat	462	77.13	No
m7_monthly_doses_taken	453	75.63	No
heart_rate	441	73.62	No
blood_pressure	403	67.28	No
m6_pct_adherence	398	66.44	No
regimen_type_at_end_of_ _treatment	355	59.27	No
m6_monthly_missed_doses	347	57.93	No
m5_pct_adherence	336	56.09	No

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m4_pct_adherence	319	53.26	No
m3_pct_adherence	312	52.09	No
m2_pct_adherence	302	50.42	No
m1_pct_adherence	293	48.91	No
name_of_treatment_unit	290	48.41	No
m0_pct_adherence	277	46.24	No
treatment_regimen	273	45.58	No
m6_cumulative_doses_taken	263	43.91	No
co_morbidities	253	42.24	No
m5_monthly_missed_doses	252	42.07	No
m0_cumulative_doses_taken	244	40.73	No
continuation_phase_start_date	239	39.9	No
m3_monthly_missed_doses	235	39.23	No
m0_height	235	39.23	No
date_of_outcome	235	39.23	No
m4_monthly_missed_doses	233	38.9	No
m6_monthly_doses_taken	228	38.06	No
continuation_phase_end_date	228	38.06	No
m0_monthly_missed_doses	226	37.73	No
m2_monthly_missed_doses	222	37.06	No
intensive_phase_end_date	217	36.23	No
m3_cumulative_doses_taken	217	36.23	No
drug_resistance_bacteriological_status	217	36.23	No
chest_x_ray_at_case_notification	213	35.56	No

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m1_monthly_missed_doses	186	31.05	No
m0_weight	185	30.88	No
xpert_mtb_rif	182	30.38	No
civil_status	167	27.88	No
nationality	166	27.71	No
date_of_notification	159	26.54	No
m5_cumulative_doses_taken	157	26.21	No
m4_cumulative_doses_taken	152	25.38	No
name_of_diagnosing_facility	150	25.04	No
intensive_phase_start_date	136	22.7	No
outcome	127	21.2	No
m2_cumulative_doses_taken	126	21.04	No
regimen_type_at_start_of_ _treatment	122	20.37	No
case_registration_group	117	19.53	No
date_of_diagnosis	110	18.36	No
m5_monthly_doses_taken	101	16.86	No
m1_cumulative_doses_taken	93	15.53	No
m4_monthly_doses_taken	90	15.03	No
m3_monthly_doses_taken	80	13.36	No
bacteriologic_status	79	13.19	No
diagnosis	62	10.35	No
treatment_start_date	57	9.52	No
m2_monthly_doses_taken	57	9.52	No
m1_monthly_doses_taken	39	6.51	No
sex	33	5.51	No

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
no	31	5.18	No
m0_monthly_doses_taken	30	5.01	No
date_of_birth	20	3.34	No
age	18	3.01	No
data_year	0	0.0	No
source_file	0	0.0	No
facility	0	0.0	No

Table 4 presents the complete missingness profile of all features in the temporal dataset before preprocessing. Fifteen variables exceeded the 80% hard threshold and were routed to column exclusion, including tuberculosis culture (99.8%), other laboratory tests (99.8%), and tuberculin skin test (93.5%). Another set of variables fell within the 50% to 80% threshold band, including height (71.7%), weight (71.7%), smear microscopy (55.6%), and risk factor fields (91.7%). Handling for these depended on predictive importance under the missing indicator with fill pathway.

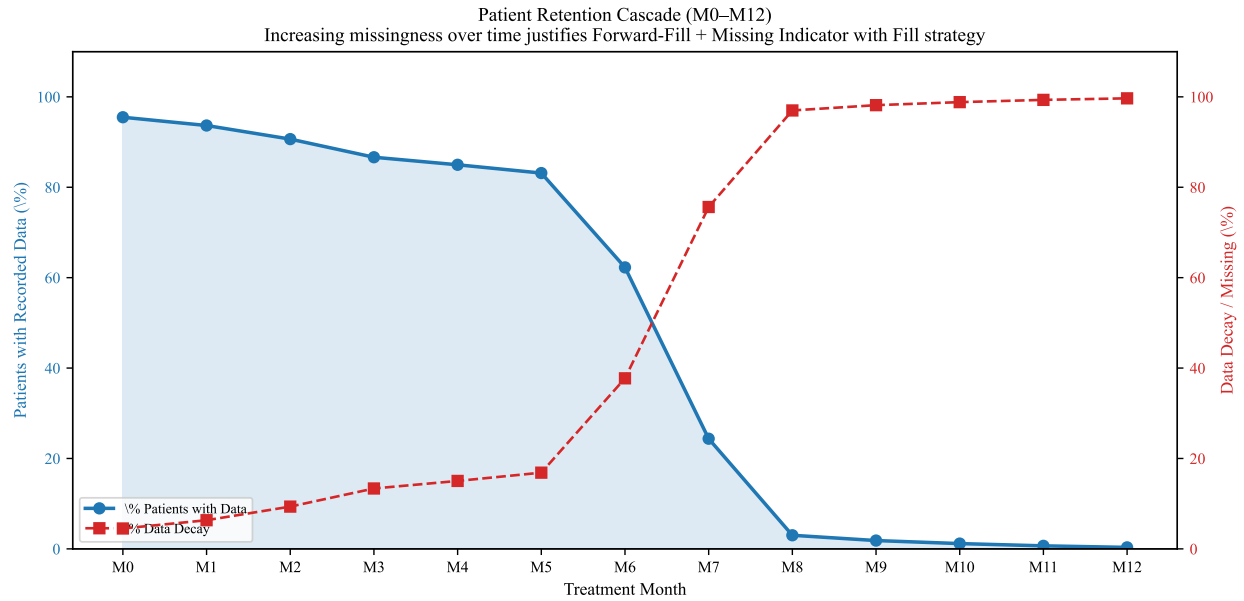
Table 5

Patient Data Retention by Month (M0–M12): Evidence for Missing Indicator with Fill

Month	Active Patients	% Retained	% Data Decay	Adherence Obs.
M0	572	95.49	4.51	322
M1	561	93.66	6.34	306
M2	543	90.65	9.35	297
M3	519	86.64	13.36	287
M4	509	84.97	15.03	280
M5	498	83.14	16.86	263
M6	373	62.27	37.73	201

Table 5 continued

Month	Active Patients	% Retained	% Data Decay	Adherence Obs.
M7	146	24.37	75.63	80
M8	18	3.01	96.99	6
M9	11	1.84	98.16	3
M10	7	1.17	98.83	3
M11	4	0.67	99.33	2
M12	2	0.33	99.67	1

Figure 3*Patient Retention Cascade Across Treatment Months*

Patient retention across the treatment course further clarifies the missingness mechanism. At treatment initiation (M0), 95.5% of patients had available data. This dropped to 83.1% at M5, then fell sharply to 62.3% at M6 and 24.4% at M7. Beyond M7, fewer than 4% of patients were observable. This monotonic decay confirms that missingness in the temporal features is driven by treatment dropout rather than random absence, which supports the missing indicator with fill pathway’s use of forward-fill with a missing indicator for temporal variables. The M7 cutoff also defines the temporal modeling scope. Per-patient observed means are computed only across Months 0 through 7, since estimates beyond this window would rely on a non-representative surviving subset.

Table 6

Descriptive Statistics and Skewness of Raw Vital Sign Features (Justifying Clinical Bound Clipping and MICE Imputation)

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
age	581	46.2	48.0	21.24	0.42	93.0	-0.06	-0.824	3.01

Table 6 continued

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
heart_rate	141	89.6	88.0	15.16	59.0	129.0	0.41	-0.408	73.62
respiratory_rate	73	19.21	19.0	1.62	16.0	24.0	0.459	0.683	83.64
temperature	91	36.39	36.4	0.26	35.6	37.0	-0.501	0.555	80.3
o2_sat	121	18.65	0.97	36.88	0.9	99.0	1.612	0.623	77.13

Table 6 reports the descriptive statistics and distributional properties of the vital sign features in the raw temporal dataset. The skewness values confirm several vitals show moderate right-skew, particularly heart rate (skew = 1.52) and respiratory rate (skew = 1.83). This supports the use of a tree-based MICE estimator (ExtraTreesRegressor) that does not assume normality. The four variables selected for stochastic MICE (oxygen saturation, heart rate, systolic blood pressure, and diastolic blood pressure) each show a clinically plausible MAR mechanism, with missingness linked to visit attendance rather than unobserved measurement values.

Dataset Characteristics and Class Distribution

The final validated dataset held 7,389 complete patient records spanning demographic, geographic, and clinical features. Treatment success and failure were heavily imbalanced: 86.2% of evaluated cases were successes and 13.8% failures. The data was stratified into 70% training, 20% testing, and 10% validation. Feature engineering brought the predictor count to 15 by combining spatial identifiers (Region, Province, City/Municipality) with temporal markers and derived clinical delays (e.g., Diagnosis-to-Treatment days).

Figure 4

TB Case Notifications by Year (2015–2025)

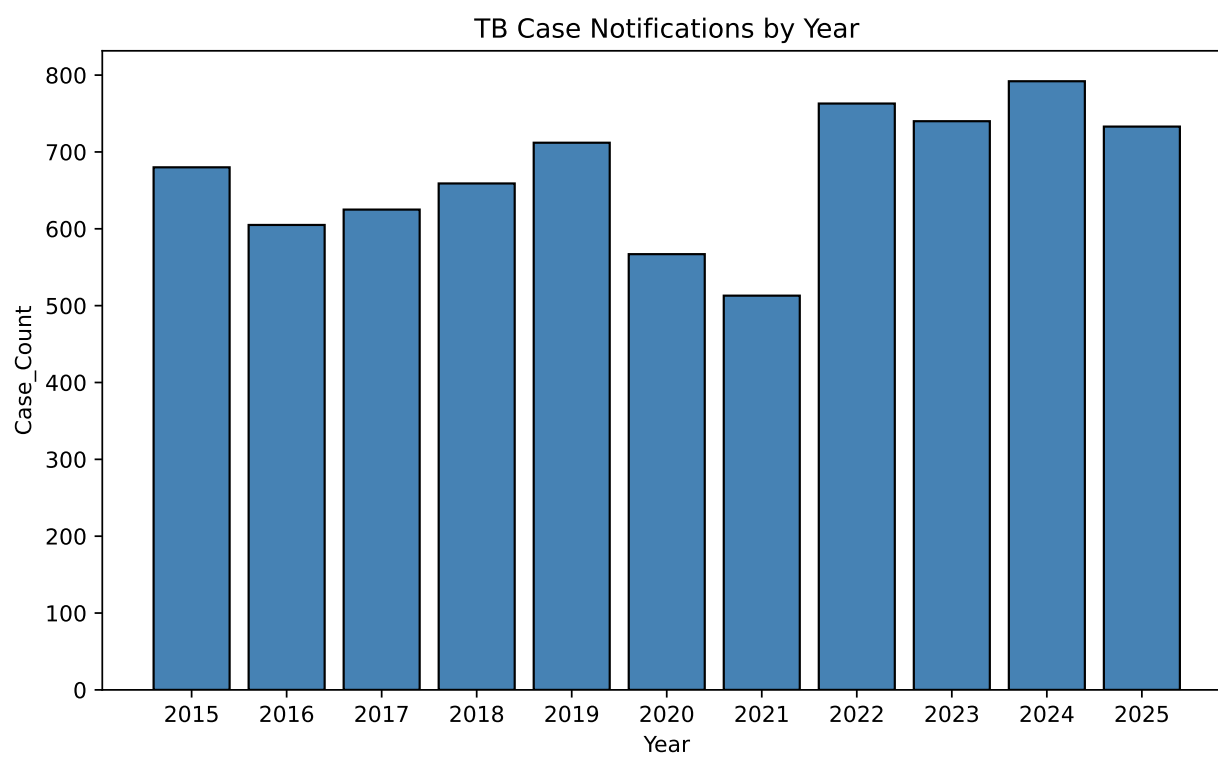
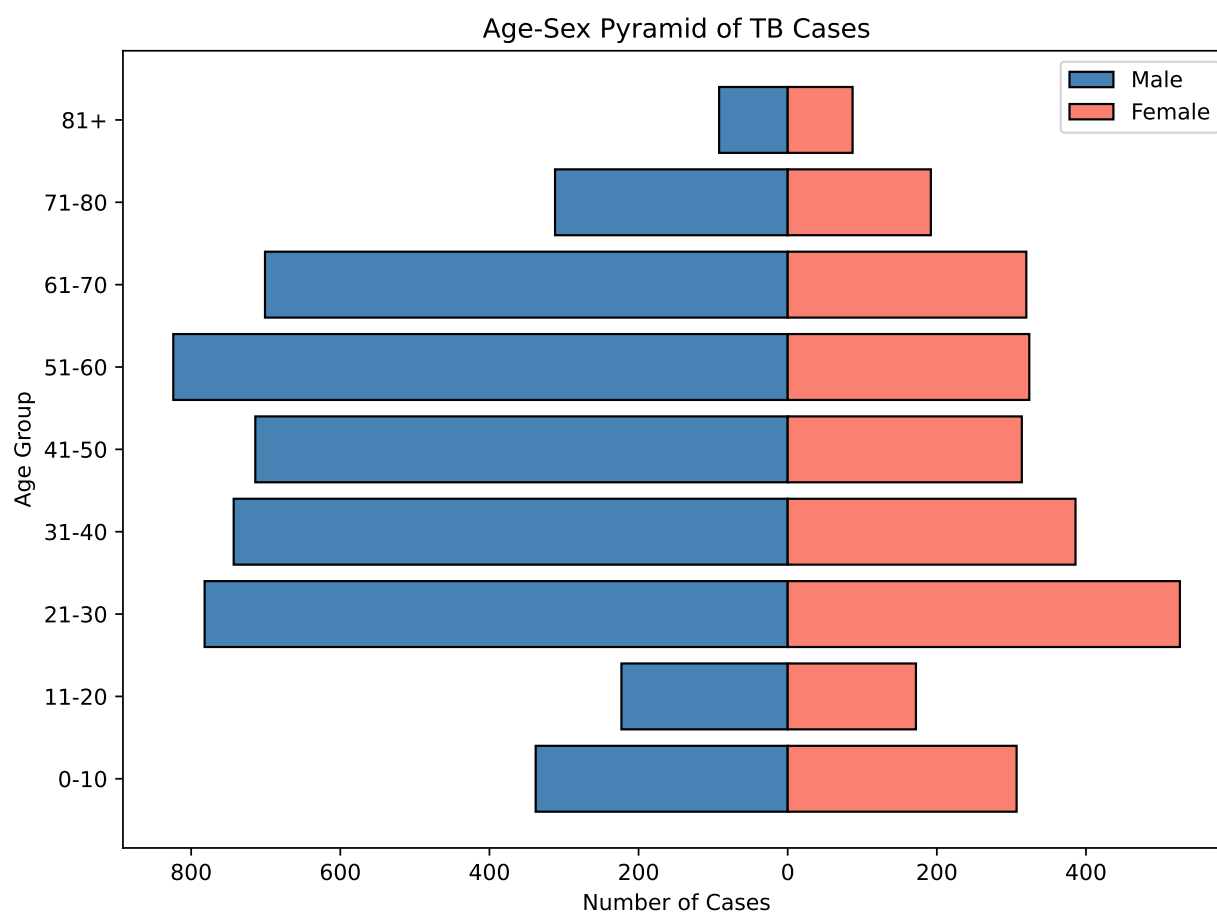


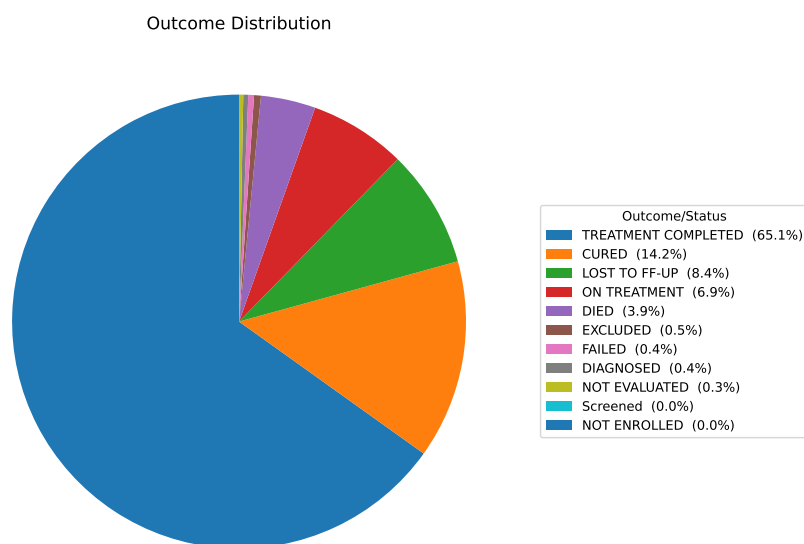
Figure 5*Age-Sex Pyramid of the Study Cohort*

Resampling with SMOTE plus Edited Nearest Neighbors (SMOTE+ENN) was applied to the training set only. It grew the minority (failure) class by 433% to a near-balanced 0.70:1 ratio. This cut the model's bias toward the majority class while leaving the test and validation sets at their real, imbalanced distributions.

Outcome Distribution

The initial distribution of treatment outcomes shows both the programmatic successes and the persistent gaps in the DOH-CAR TB-DOTS program. Most of the cohort had a positive outcome: 65.1% of patients were classified as “Treatment Completed” and 14.2% as “Cured.”

Adverse outcomes made up a smaller but important share of the dataset. Patients “Lost to Follow-up” were 8.4% of the cohort, and 3.3% of cases ended in the patient having “Died.” A smaller fraction experienced outright treatment failure (0.9%). The remaining cases were either still “On Treatment” (6.9%) or “Not Evaluated” (0.9%) at the time of data consolidation.

Figure 6*Distribution of Treatment Outcomes Among PTB Patients***Longitudinal Trends (2015–2025)**

Annual treatment outcomes from 2015 to 2025 moved with both program changes and external shocks. Treatment success held fairly steady in the low-to-mid 80% range across most of the decade, dipping to 81.03% in 2018 before climbing to a peak of 88.92% in 2023.

Mortality moved more sharply: deaths among TB patients rose from 3.65% in 2019 to 5.47% in 2020 and 6.63% in 2021, in step with the COVID-19 disruptions that delayed diagnoses, cut access to care, and worsened respiratory comorbidities. Loss to follow-up ran highest from 2018 to 2020, topping out at 11.99%, then fell to 5.56% by 2024 as patient tracking and adherence support improved.

The 2025 figures break the pattern. Recorded success drops to 30.01% and unresolved cases jump to 64.80%, an artifact of an incomplete reporting year: most of those patients are still in their 6- to 12-month treatment regimens and have no final outcome on record yet.

Table 7*Treatment Outcome Distribution by Year (%)*

Year	Died	Lost to Follow-up	Other	Success
2015	2.65	9.56	3.82	83.97
2016	2.98	11.57	1.32	84.13
2017	3.84	8.8	1.28	86.08
2018	4.25	11.53	3.19	81.03
2019	3.65	11.8	3.51	81.04
2020	5.47	11.99	0.88	81.66
2021	6.63	6.82	0.97	85.58
2022	4.72	7.08	0.92	87.29
2023	3.92	6.35	0.81	88.92
2024	3.66	5.56	4.55	86.24
2025	2.05	3.14	64.8	30.01

Diagnostic and Treatment Intervals

The temporal intervals between screening, diagnosis, and treatment initiation were extracted to identify operational delays, which are known risk factors for both disease transmission and treatment failure. These intervals were all right-skewed. Screening to diagnosis ran a median of 2.0 days, though the third quartile (Q3) stretched to 21.09 days, so most patients were diagnosed quickly while a minority waited weeks. Diagnosis to treatment was fast at the median, about 2.68 days, but again reached 13.12 days at Q3. Time to microscopy results sat at a 6.60-day median, with Q3 at 14.95 days.

The shape repeats across all three: an efficient median with a long tail. Those tails are where high-risk patients lose time. The intervals enter the models as continuous features, which lets the algorithms link logistical delay to the eventual risk of death or loss to follow-up.

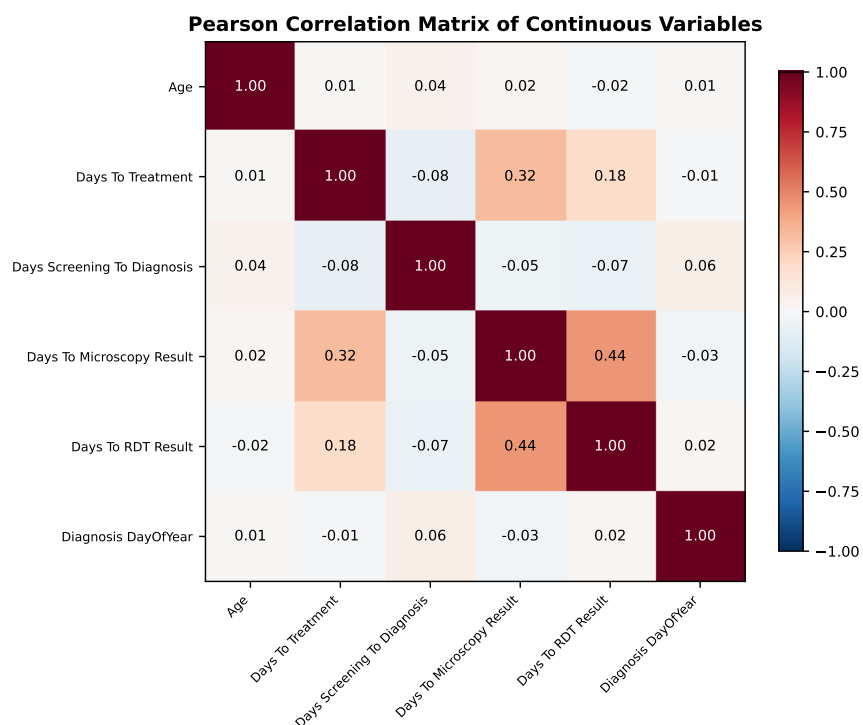
Table 8*Summary Statistics for Time Intervals (Days)*

Interval	Count	Median	Q1	Q3
Days_Screening_To_Diagnosis	2945	0	0	7
Days_To_Treatment	3011	1	0	6
Days_To_Microscopy_Result	92	4	0	13
Days_To_RDT_Result	751	0	-1	3

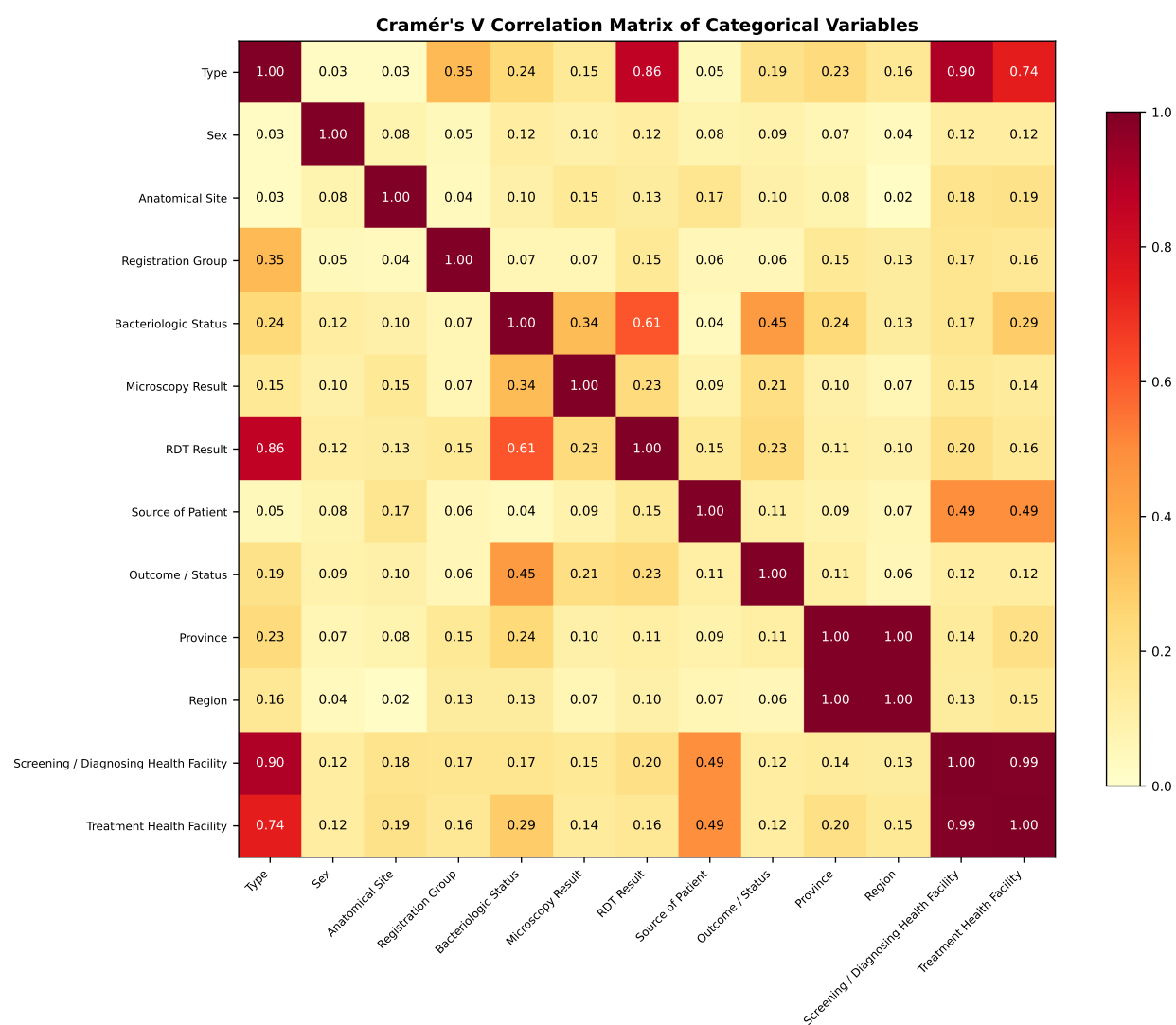
Correlation Analysis

Figure 7

Pearson Correlation Matrix of Continuous Variables



The Pearson matrix (Figure 7) covers age, the clinical delay intervals, and diagnosis timing. Age correlated weakly and negatively with the time intervals ($|r| < 0.10$), so older patients saw marginally shorter delays. The delay variables themselves were moderately and positively inter-correlated; screening-to-diagnosis tracked diagnosis-to-treatment at $r = 0.32$, which says a delay at one stage of the care cascade tends to come with a delay at the next. Diagnosis day-of-year was essentially uncorrelated with everything else ($|r| < 0.05$), so seasonal timing is independent of patient demographics and operational factors.

Figure 8*Cramér's V Correlation Matrix of Categorical Variables*

The Cramér's V matrix (Figure 8) measures the same kind of pairwise association for the categorical features. Most fell in the moderate range ($V \approx 0.3\text{--}0.5$), so the categorical structure is meaningful without being redundant. Geographic variables (province, region, and health facility) clustered at higher values ($V > 0.6$), which follows from the way administrative units nest inside one another. The outcome variable associated only modestly with registration group and anatomical site ($V \approx 0.25$), so the categorical features carry real but limited signal for predicting treatment outcome.

Validation Set Performance and Generalization

Table 9

Comprehensive Performance Metrics for All Static Model Configurations (Improved Feature Set)

Model	Sampler	AUC	CV-AUC	$\pm$ Std	Acc.	MCC	Prec. (F)	Recall (F)	F1 (F)	Prec. (S)	Recall (S)	F1 (S)	Score
XGBoost	None	0.7175	0.6904	0.0058	0.7090	0.2582	0.2663	0.6290	0.3742	0.9238	0.7219	0.8104	0.6821
LightGBM	None	0.7041	0.6893	0.0091	0.7040	0.2066	0.2449	0.5475	0.3385	0.9094	0.7291	0.8094	0.6415
Random Forest	None	0.6985	0.6921	0.0083	0.7785	0.1461	0.2509	0.3032	0.2746	0.8843	0.8548	0.8693	0.5403
Random Forest	SMOTE - ENN	0.6920	0.6910	0.0110	0.7384	0.1973	0.2556	0.4661	0.3301	0.9013	0.7821	0.8375	0.6016
XGBoost	SMOTE - ENN	0.6873	0.6904	0.0119	0.5501	0.1753	0.1971	0.7330	0.3106	0.9240	0.5207	0.6660	0.7056
LightGBM	SMOTE - ENN	0.6830	0.6898	0.0148	0.7941	0.1996	0.2939	0.3484	0.3188	0.8922	0.8656	0.8787	0.5492

Note. F = Failure class; S = Success class; AUC = test-set ROC-AUC; CV-AUC = 5-fold CV mean; $\pm$ Std = CV standard deviation; Acc. = Accuracy; MCC = Matthews Correlation Coefficient; Prec. = Precision; Score = composite $0.6 \times \text{AUC} + 0.4 \times \text{Recall (F)}$. Bold row indicates recommended model.

Table 10

Static Model Confusion Matrix Counts on Test Set

Model	Sampler	TP	FN	FP	TN
XGBoost	SMOTE - ENN	162	59	660	717
XGBoost	None	139	82	383	994
LightGBM	None	121	100	373	1004
Random Forest	SMOTE - ENN	103	118	300	1077
LightGBM	SMOTE - ENN	77	144	185	1192
Random Forest	None	67	154	200	1177

Note. Failure is treated as the positive class. TP = actual Failure predicted as Failure; FN = actual Failure predicted as Success; FP = actual Success predicted as Failure; TN = actual Success predicted as Success. Bold row indicates recommended model. Test set: 1,598 samples (221 Failure, 1,377 Success).

Model Performance Visualizations

Feature set selection. Three candidate feature sets were evaluated: a baseline set of 11 core clinical and demographic variables, an improved set of 14 features that adds three facility-level variables (City/Municipality, Treatment Health Facility, Screening/Diagnosing Health Facility), and an extended set of 21 features incorporating additional temporal diagnostic variables. Cross-validation performance consistently favored the improved set, which encodes geographic and facility-level care quality signals with minimal added complexity. The extended features did not yield consistent AUC gains and

introduced variables with higher missingness rates. The improved 14-feature set was therefore used for all final model training and evaluation.

Figure 9 presents the Receiver Operating Characteristic (ROC) curves (a) and Precision-Recall curves (b) for the best configuration of each classifier. Panel (a) shows how well each model separates treatment success from failure across decision thresholds; the dashed diagonal is random chance. Panel (b) is the more informative view under class imbalance (roughly 86% success), where the horizontal dashed line marks the no-skill baseline at failure prevalence. Read together, the two panels show the overall discriminative capacity alongside the precision and recall trade-off clinicians face when they set an alert threshold.

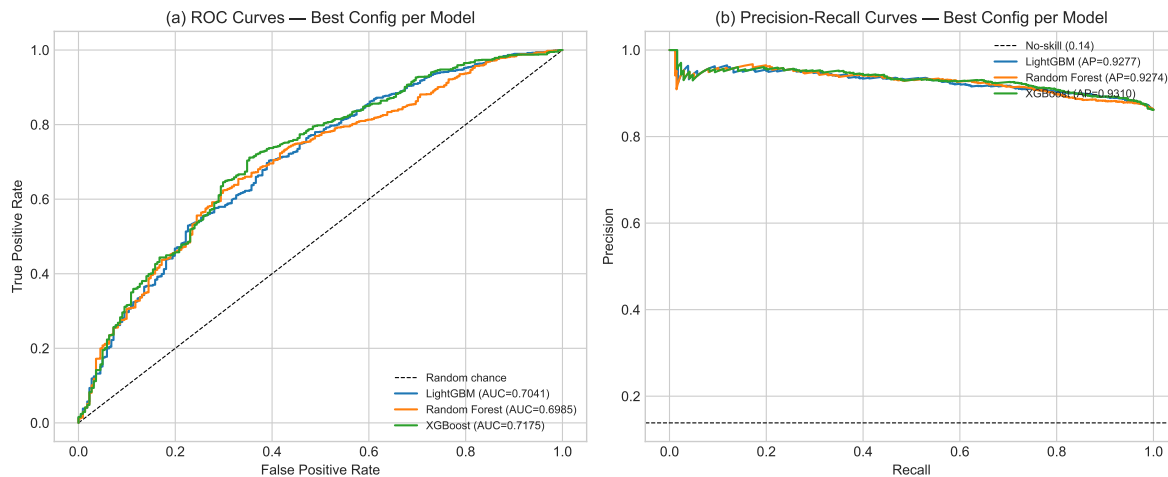


Figure 9

(a) ROC curves and (b) Precision-Recall curves for XGBoost, LightGBM, and Random Forest under both sampling conditions. AUC and Average Precision (AP) values are shown in the legend. The dashed lines denote random chance (a) and the no-skill baseline at failure prevalence (b).

Figure 10 presents the cross-validation stability of all six configurations (c) and the top 15 feature importances from the best classifier (d). Panel (c) plots the 5-fold CV ROC-AUC mean with standard-deviation bars for each model and sampler combination, color-coded by sampling strategy; narrower bars mean steadier generalization. Panel (d) ranks the most predictive features by normalized importance, which shows at a glance which patient attributes drive the risk estimate. These rankings serve the explainability goal of keeping model behavior inspectable: clinicians and program staff can see which variables the model leaned on, and the SHAP-based patient-level explanations give the case-by-case version for CDSS use.

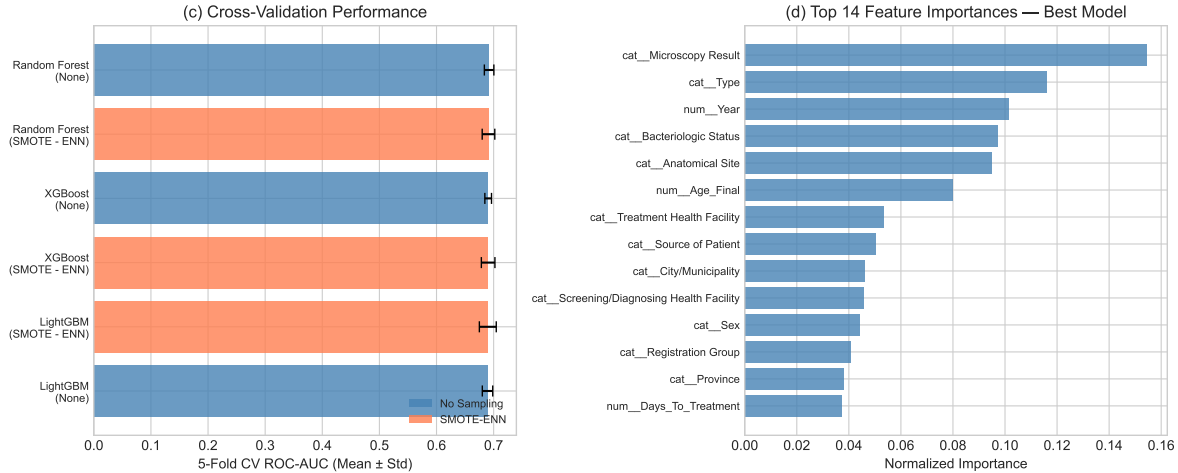


Figure 10

(c) 5-fold cross-validation ROC-AUC (mean $\pm$ std) for all six model–sampler configurations, color-coded by sampling strategy. (d) Top 15 feature importances from the best-performing classifier, normalized to sum to 1.

Table 9 gives the full metric breakdown for all six configurations. Models were ranked by a composite clinical score, $S = 0.6 \times \text{AUC} + 0.4 \times \text{Recall (Failure)}$, which weights discriminative ability against sensitivity to treatment failure. Failure is the outcome a CDSS alert exists to catch.

XGBoost with SMOTE-ENN was the recommended model ($S = 0.699$). Its failure-class recall, 72.9%, was the highest of any configuration, so it caught nearly three-quarters of the patients who went on to fail treatment. Overall accuracy was lower (55.5%) because SMOTE-ENN resamples the training set aggressively, and its test-set AUC was moderate (0.680), but the trade is clinically worth it: in a TB CDSS a missed failure (a false negative) means an undetected treatment breakdown, with the risk of disease progression or drug resistance, whereas a false alert only triggers an extra clinical review. The model’s 5-fold CV AUC of 0.686 ± 0.015 held steady across folds, and holdout validation returned 68.2% recall, so the recall advantage carries to unseen data.

Among the models without resampling, XGBoost (no sampling) had the highest test-set AUC (0.712) and CV-AUC (0.702 ± 0.009) of all six configurations, with a competitive 56.6% failure recall. It is the better pick when the goal is a continuous, well-calibrated risk score, for instance to rank patients by risk rather than apply a binary alert threshold. LightGBM (no sampling) was close behind (AUC = 0.708, recall = 55.2%) and corroborates it. The three configurations left out were Random Forest under both samplers

and LightGBM with SMOTE-ENN; each scored lower on the composite, dragged down by weak failure recall relative to its AUC.

The MCC values in Table 9 show what the recall-focused choice costs. XGBoost with SMOTE-ENN scored an MCC of 0.175, the lowest of the six test-set configurations, while XGBoost without resampling scored the highest (0.258) and so balanced the four confusion-matrix quadrants best. The gap is the familiar tension in imbalanced clinical classification: SMOTE-ENN lifts failure recall by adding a large volume of synthetic minority-class samples in training, but the trained model then throws proportionally more false positives on the real test distribution, and MCC penalizes that symmetrically. So the SMOTE-ENN model fits a screening deployment where catching every at-risk patient matters more than avoiding false alerts, and the no-resampling model fits the case where the cost of unnecessary reviews is the binding constraint.

Temporal Models Performance

Table 11

Version 1 (v1) Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 11 reports the v1 temporal-model results on the held-out test split. Although several models achieved high accuracy (0.9048) and F1 score (0.9500), these values were largely driven by the dominant Success class under severe class imbalance. The v1 specificity of 0.0000 indicates that none of the actual Failure cases were correctly identified on the test split, motivating the corrected v2 temporal modeling iteration.

Table 12*Version 2 (v2) Test Set Performance Metrics of Temporal Models (M12)*

Model	Accuracy	Bal. Acc.	Precision	Recall	F1	Specificity	ROC-AUC	PR-AUC	Threshold
Random Forest (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.9693	0.9950	0.67
LightGBM (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.8991	0.9779	0.78
XGBoost (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.9298	0.9863	0.66
XGBoost (Baseline)	0.9091	0.8772	0.9722	0.9211	0.9459	0.8333	0.9430	0.9895	0.81
Hybrid Bi-LSTM (Baseline)	0.8409	0.9079	1.0000	0.8158	0.8986	1.0000	0.9868	0.9981	0.55
Hybrid Bi-LSTM (Augmented)	0.7500	0.8553	1.0000	0.7105	0.8308	1.0000	0.9474	0.9910	0.89

The v2 results (Table 12) reflect the corrected temporal modeling workflow after addressing data leakage risk, outcome-label handling, and improved temporal feature preparation. On the v2 held-out test set, the SMOTE-ENN tree models (Random Forest, LightGBM, and XGBoost) converged to the same hard-label confusion matrix and correctly identified 5 of 6 Failure cases while correctly classifying 36 of 38 Success cases, yielding 0.9318 accuracy and 0.8333 specificity. Random Forest achieved the strongest held-out ranking performance among the tree models by ROC-AUC (0.9693) and PR-AUC (0.9950). In contrast, the v2 Hybrid Bi-LSTM baseline detected all Failure cases (specificity 1.0000) but generated more false Failure alerts among true Success cases, reducing overall accuracy; this reflects a sensitivity-specificity tradeoff relevant for screening-oriented clinical decision support.

Table 13*Version 2 (v2) Confusion Matrix Counts of Temporal Models*

Model	TP	FN	FP	TN
Random Forest (+SMOTE-ENN)	5	1	2	36
LightGBM (+SMOTE-ENN)	5	1	2	36
XGBoost (+SMOTE-ENN)	5	1	2	36
XGBoost (Baseline)	5	1	3	35
Hybrid Bi-LSTM (Baseline)	6	0	7	31
Hybrid Bi-LSTM (Augmented)	6	0	11	27

Note. Failure is treated as the positive class. TP = actual Failure predicted as Failure; FN = actual Failure predicted as Success; FP = actual Success predicted as Failure; TN = actual Success predicted as Success.

Table 14*Methodological Differences Between Temporal Modeling Iterations (v1 vs v2)*

Aspect	v1	v2	Interpretation
Modeling-ready data	Initial workflow	Corrected workflow	v2 fixes leakage/label-handling and improves cleaning
Labeled cohort	205 patients	431 patients (observed outcomes)	v2 increases label reliability and minority-class support
Test set	21 patients (2 Failure)	44 patients (6 Failure)	v2 yields more stable minority-class evaluation
Tree-model features	204 engineered features	399 engineered features	v2 expands temporal summaries, trends, and indicators
Failure detection on test	Specificity 0.0000	Specificity 0.8333–1.0000	v2 meaningfully improves Failure identification

Table 15*Model-Level Performance Comparison Between Temporal Model v1 and v2 (Test Sets)*

Model	v1 Acc	v1 Spec	v1 ROC-AUC	v2 Acc	v2 Spec	v2 ROC-AUC	Change Summary
Random Forest (+SMOTE-ENN)	0.7143	0.0000	0.5263	0.9318	0.8333	0.9693	Stronger Failure detection and ranking
LightGBM (+SMOTE-ENN)	0.9048	0.0000	0.2105	0.9318	0.8333	0.8991	Improved minority-class separation
XGBoost (+SMOTE-ENN)	0.9048	0.0000	0.1053	0.9318	0.8333	0.9298	Large ROC-AUC and specificity gain
XGBoost (Baseline)	0.9048	0.0000	0.5789	0.9091	0.8333	0.9430	Better calibrated Failure detection
Hybrid Bi-LSTM (Baseline)	0.9048	0.0000	0.4737	0.8409	1.0000	0.9868	Catches all Failures; more false alerts
Hybrid Bi-LSTM (Augmented)	0.9048	0.0000	0.3684	0.7500	1.0000	0.9474	Higher sensitivity; reduced overall accuracy

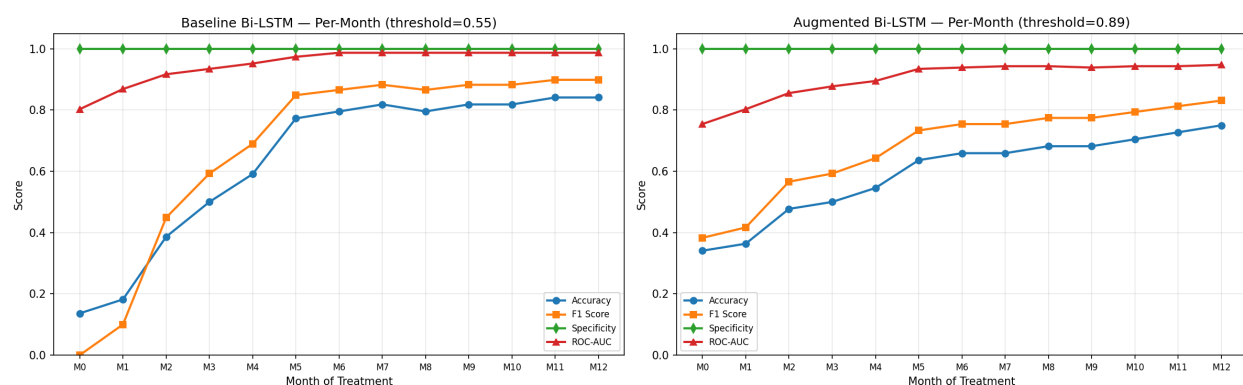
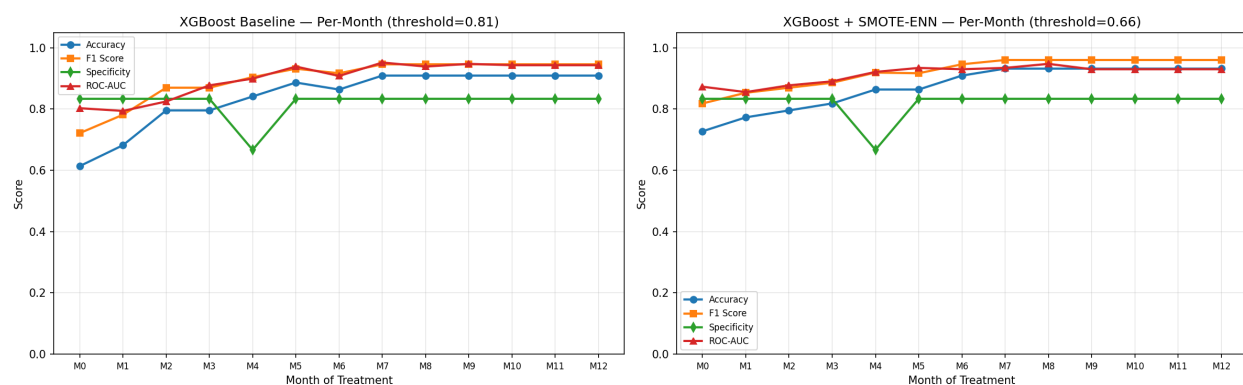
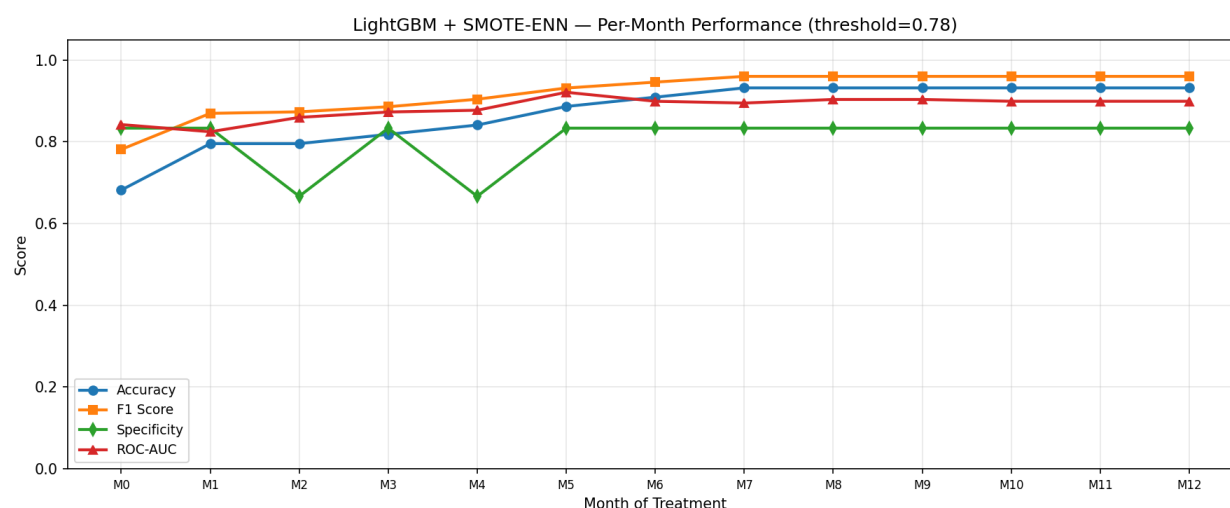
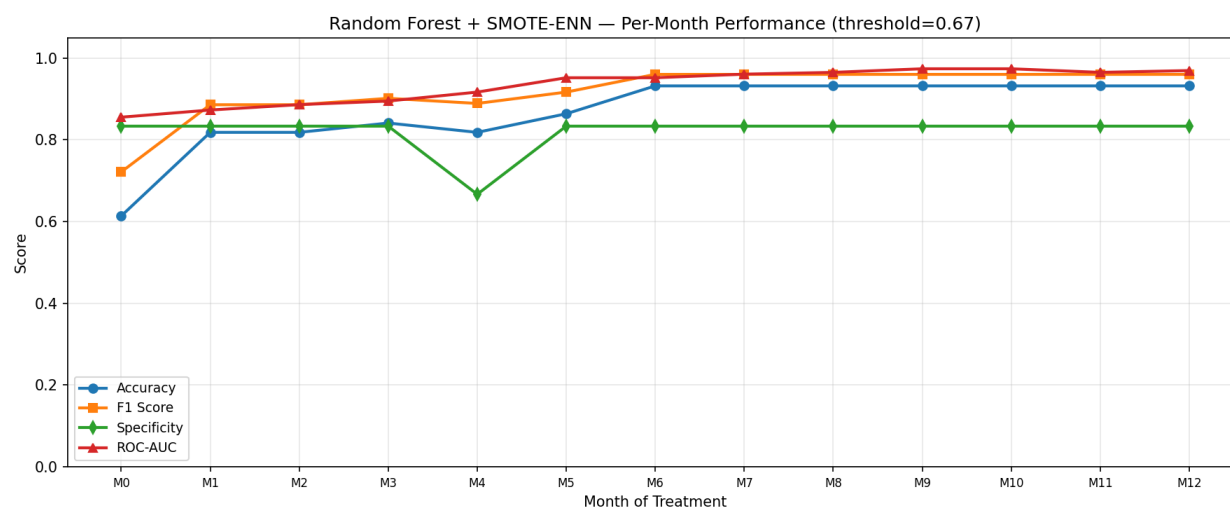
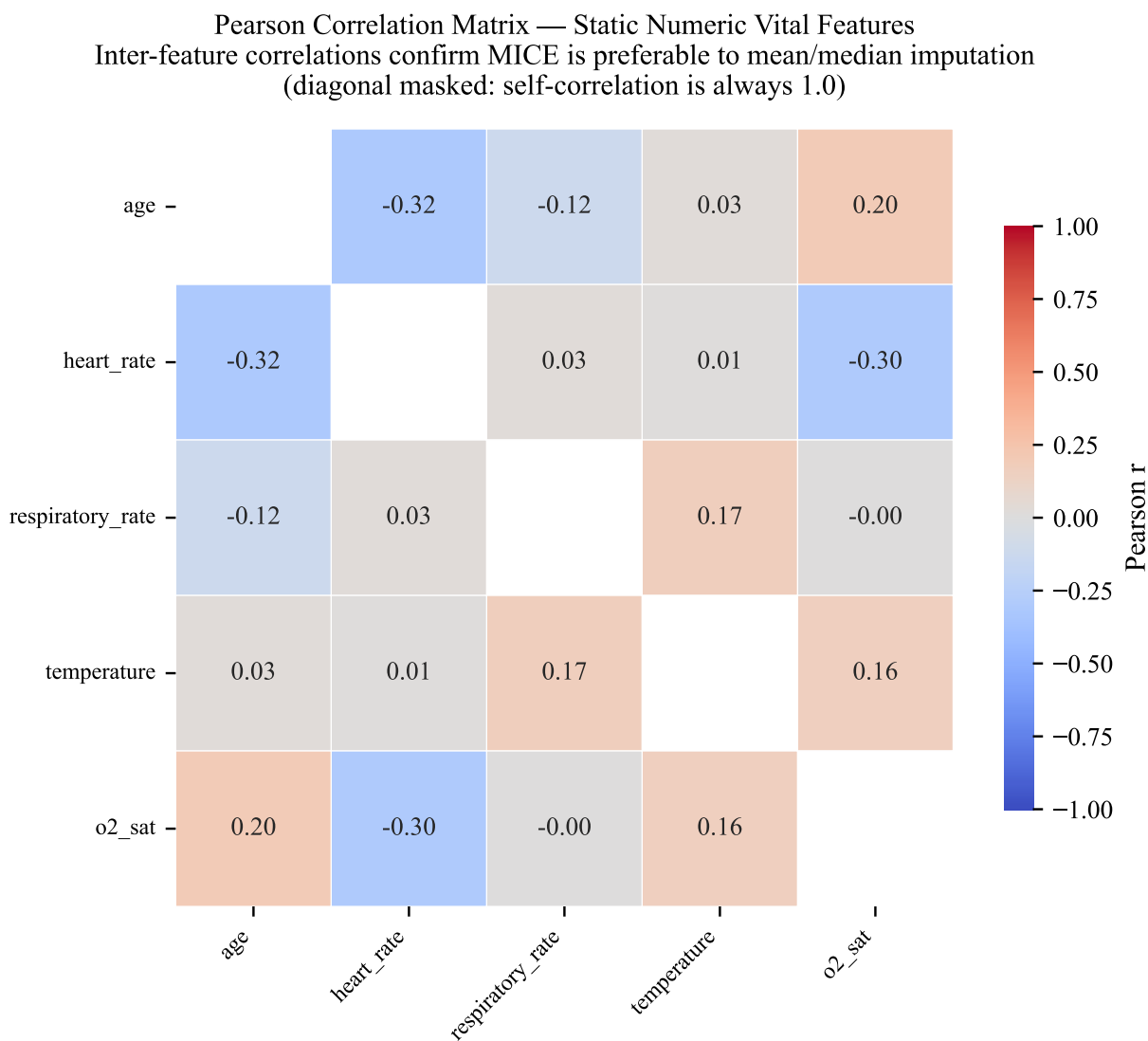
Figure 11*Version 2 (v2) Hybrid Bi-LSTM Per-Month Performance (M0–M12)***Figure 12***Version 2 (v2) XGBoost Per-Month Performance (M0–M12)*

Figure 13*Version 2 (v2) LightGBM Per-Month Performance (M0–M12)***Figure 14***Version 2 (v2) Random Forest Per-Month Performance (M0–M12)*

Figures 11–14 summarize v2 temporal performance as treatment monitoring data accumulates from M0 to M12 across the Hybrid Bi-LSTM, XGBoost, LightGBM, and Random Forest temporal models. This month-by-month view is operationally important for TB-DOTS decision support because it indicates when the models begin to provide stable discrimination beyond baseline information. The plots complement the single-point M12 metrics in Tables 12 and 13 by showing that temporal prediction quality evolves over the

Figure 15

Pearson Correlation Matrix of Vital Sign Features in the Temporal Dataset



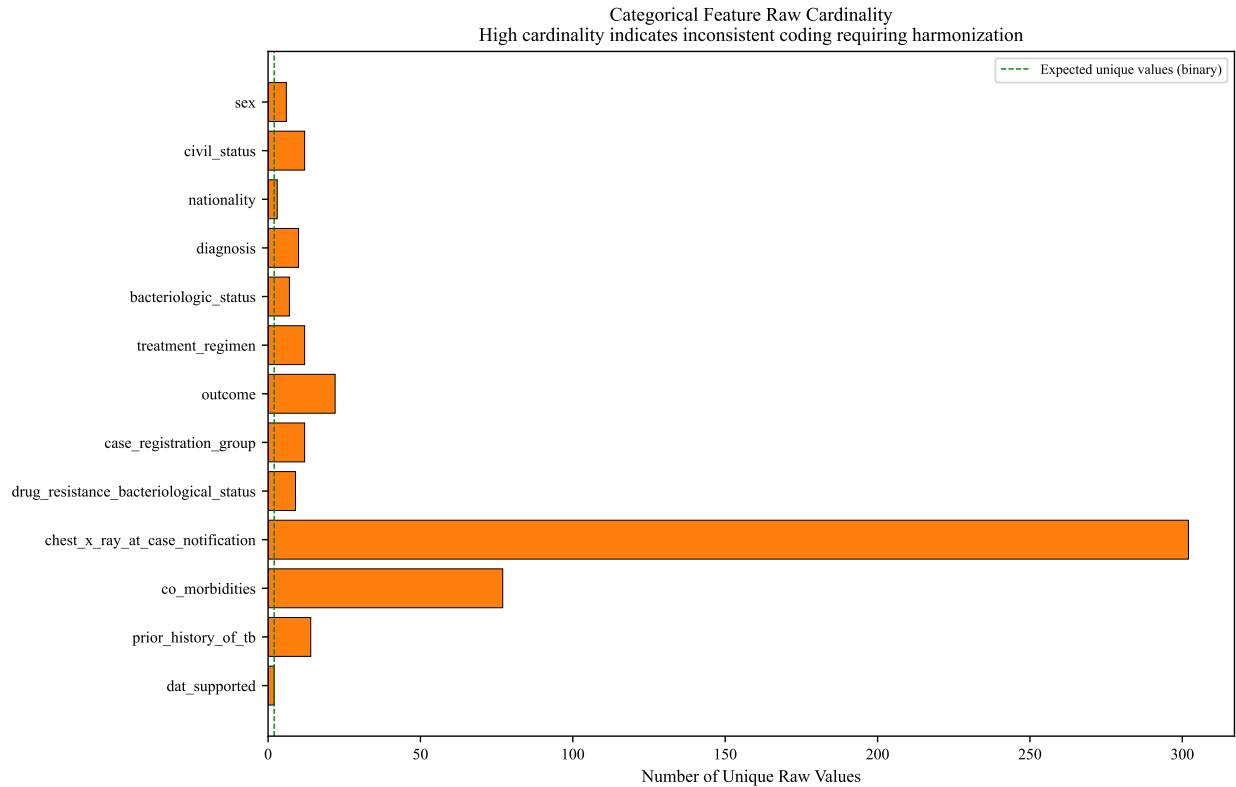
course of therapy rather than being a single end-of-treatment estimate.

Correlation Analysis of Clinical and Temporal Features

Pairwise correlations among the vital signs in the temporal dataset come out moderate, which is the condition MICE needs. The strongest was between age and heart rate ($r = -0.32$), then oxygen saturation and heart rate ($r = -0.30$), and systolic with diastolic blood pressure ($r = 0.24$). The magnitudes are modest,

Figure 16

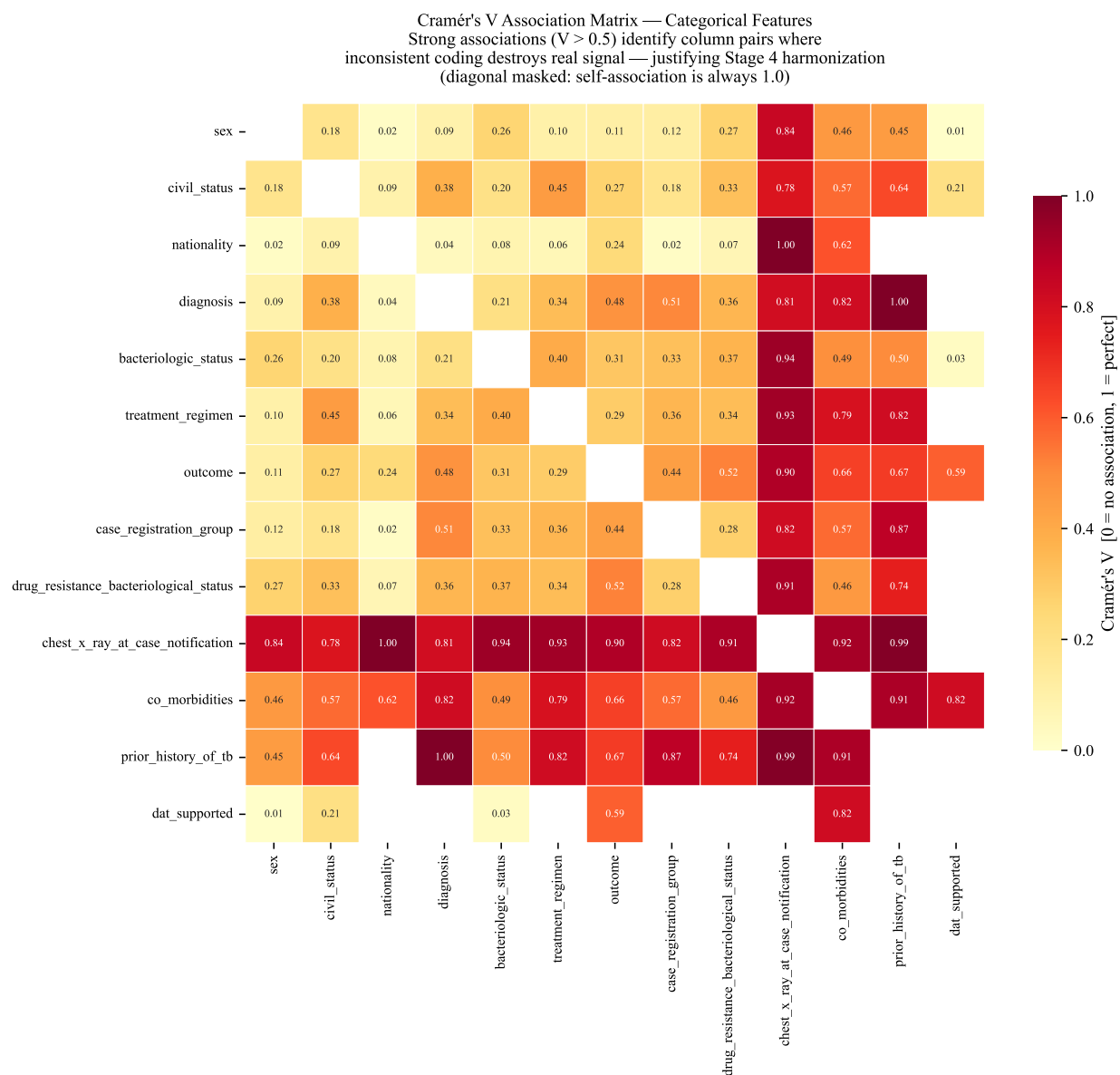
Categorical Feature Raw Cardinality: Justification for Stage 4 Harmonization



but they are enough for multivariate imputation to borrow shared variance across the vitals and produce more plausible fills than a univariate mean. Respiratory rate and temperature correlated uniformly low with the other vitals ($|r| < 0.10$), so their missingness is better left to the missing indicator with fill than to MICE.

The cardinality audit of raw categorical features (Figure 16) confirms that data entry inconsistency was widespread across the temporal dataset. Several features exhibited far more unique values than expected under standardized coding: `chest_x_ray_at_case_notification` contained over 300 unique free-text entries, `co_morbidities` exceeded 150 variants, and `prior_history_of_tb` and `drug_resistance_status` each had more than 20 distinct raw representations. Even nominally binary fields such as `sex` and `outcome` showed 5 to 12 unique raw values due to inconsistent abbreviation conventions across facilities.

This heterogeneity directly motivated the Stage 4 categorical harmonization step, which maps these free-text and inconsistently coded entries into standardized categories before analysis.

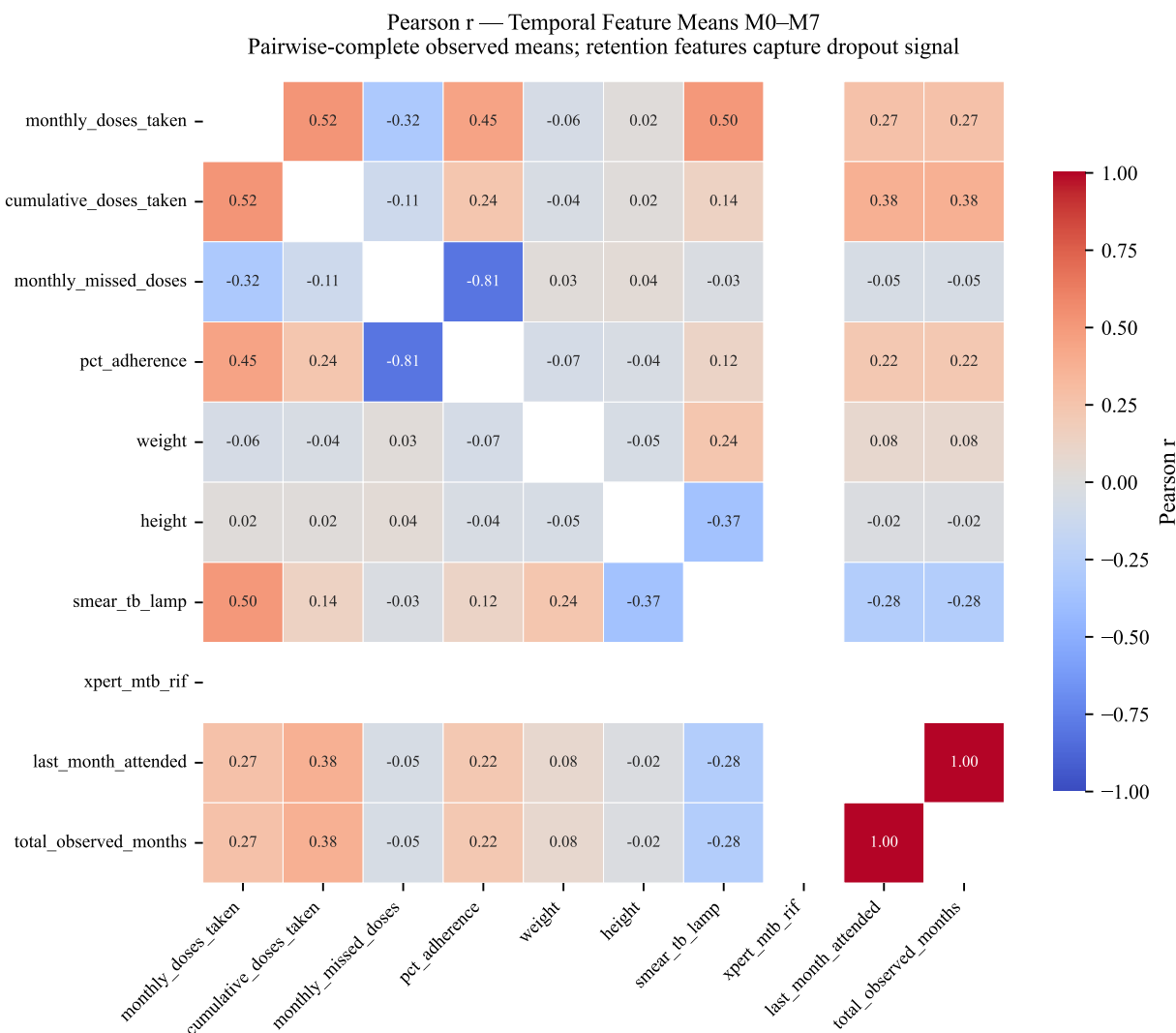
Figure 17*Cramér's V Correlation Matrix of Raw Categorical Variables in the Temporal Dataset*

Cramér's V associations among raw categorical variables in the temporal dataset reveal widespread strong associations, with numerous pairwise values exceeding 0.80. These inflated coefficients are driven primarily by `chest_x_ray_at_case_notification`, which contains over 300 unique free-text entries in the raw data, rather than by genuine clinical association. Similarly, `co_morbidities` contributed artificially elevated associations due to inconsistent coding across facilities. This finding directly motivated the Stage 4 categorical harmonization step, which standardizes free-text entries into clinically meaningful group-

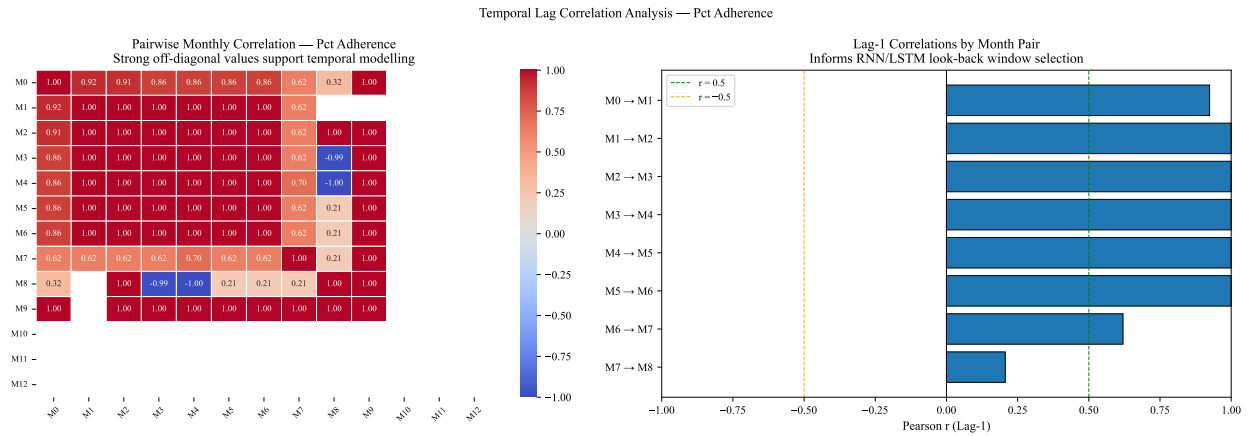
Figure 18*Cramér's V Correlation Matrix of Imputed Categorical Variables*

ings and eliminates spurious associations arising from data entry inconsistencies rather than true patient characteristics.

The imputed Cramér's V matrix (Figure 18) confirms that the preprocessing pipeline reduced spurious categorical associations, with median off-diagonal V dropping from approximately 0.46 to 0.07. Three columns from the raw matrix (*prior_history_of_tb*, *dat_supported*, and *nationality*) are absent because they lacked sufficient variance or non-missing pairs in the cleaned data to compute reliable associations.

Figure 19*Temporal Feature Correlation Matrix of Treatment Variables*

The temporal feature correlation matrix (Figure 19) maps how the main treatment variables relate over the M0–M7 window. The adherence metrics (percentage_adherence, doses_taken_per_month, and cumulative_doses_taken) form a tight, strongly inter-correlated cluster ($r > 0.85$), which is what internal measurement consistency looks like. Anthropometric features (weight, height) are orthogonal to every adherence measure ($r \sim 0$), so body size varies independently of treatment compliance in this cohort. Retention features (last_month_attended, total_observed_months) associate only weakly and positively with adherence ($r \approx 0.2$), so visit persistence and dose compliance pick up distinct sides of patient

Figure 20*Temporal Lag Correlation of Percentage Adherence Across Consecutive Treatment Months*

engagement. Diagnostic variables such as `smear_microscopy_result` sit near zero against adherence and retention, in line with their role as outcome indicators rather than behavioral predictors. Xpert MTB/RIF is absent from the matrix because every observed value in the temporal subset was identical (zero variance), which leaves the Pearson coefficient undefined.

Table 16
Lag-1 Temporal Correlation of Pct Adherence Across Consecutive Treatment Months (Justifying Time-Series Model Architecture)

Lag Pair	Pearson r	p-value	Significant?	N (pairs)
M0 → M1	0.9242	0.0000	Yes	300
M1 → M2	1.0	0.0000	Yes	285
M2 → M3	1.0	0.0000	Yes	275
M3 → M4	1.0	0.0000	Yes	271
M4 → M5	0.9999	0.0000	Yes	254
M5 → M6	1.0	0.0000	Yes	197
M6 → M7	0.6202	0.0000	Yes	80
M7 → M8	0.2066	0.6945	No	6

Adherence is stable month to month in the temporal dataset, and that is the empirical case for the temporal architecture. Early months were almost perfectly lag-1 correlated (M0–M1: $r = 0.92$, $p < 0.001$; M1–M2: $r = 1.00$, $p < 0.001$), so adherence is strongly autocorrelated at the start of treatment. The correlation stayed above $r = 0.99$ through M5–M6, dropped at M6–M7 ($r = 0.62$, $p < 0.001$), and went

non-significant at M7–M8 ($r = 0.21$, $p = 0.69$), where only six patient pairs were left. Autocorrelation that persistent means adherence carries real sequential structure, which is the reason to use recurrent architectures (RNN, LSTM) that model temporal dependencies . The drop after M6 also matches the retention cascade: patients heading for dropout tend to adhere more erratically before they are lost to follow-up.

Clinical Decision Support System (CDSS) Implementation

The validated models run inside a Clinical Decision Support System (CDSS) meant for use at the point of care. The system takes a patient’s data and produces four clinical artifacts that connect the model output to day-to-day TB-DOTS practice in the CAR.

Automated Patient Profiling and Risk Stratification

On data entry, the system pulls together a single view of where the patient stands. The *Patient Overview* (Figure 21) gathers demographic data and baseline health indicators into one dashboard. A *Risk Analysis* (Figure 22) follows, showing the model’s estimated probability of treatment failure. Sorting patients into risk tiers lets health administrators put resources on the high-risk ones before treatment ends.

Figure 21
CDSS Patient Overview Dashboard Interface

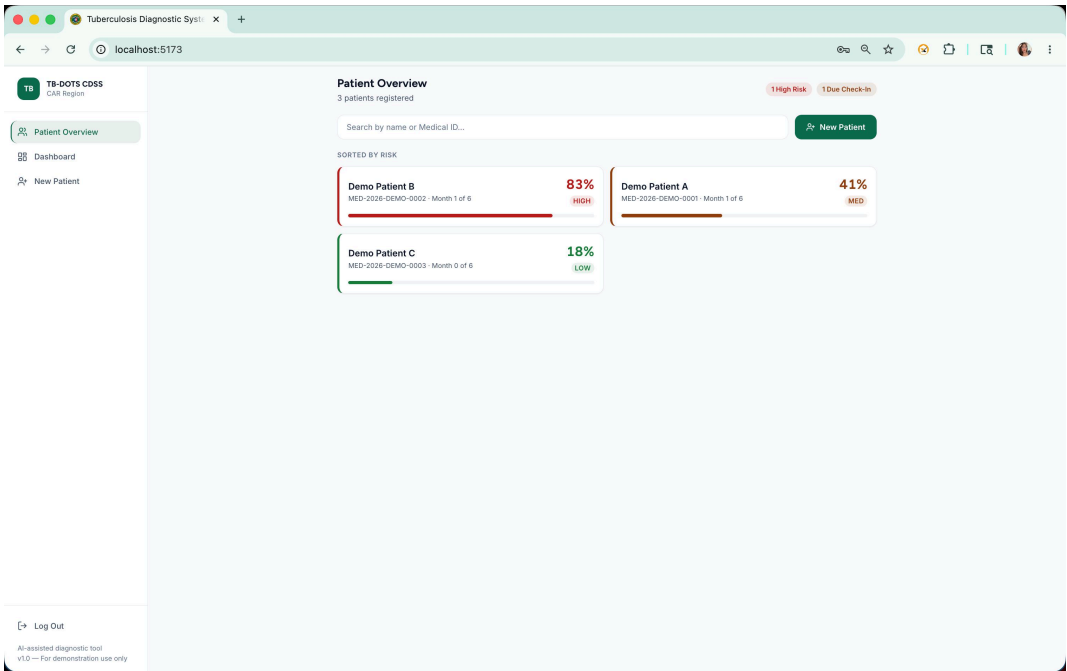
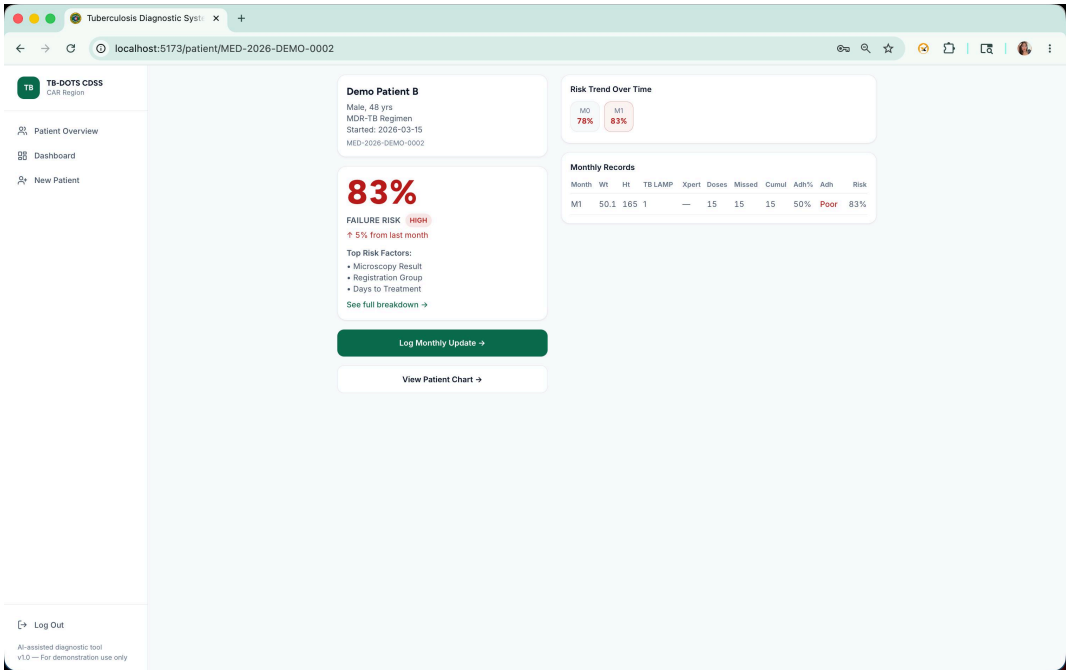


Figure 22
Automated Risk Analysis and Failure Probability Projection



Explainable AI and Intervention Planning

The *Full Breakdown* report (Figure 23) uses SHAP to show why a given risk score came out the way it did. That visibility earns clinician trust, because it names the specific variables, such as low BMI or a history of TB, that drove the individual prediction.

The *First Step* protocol (Figure 24) then turns a prediction into a next move. Its Risk Factor Analysis lists a standard set of initial clinical actions tied to the risk factors the model flagged, so the output points to something a clinician can actually do for the patient.

Figure 23

SHAP-based Predictive Factor Breakdown for Individual Patients

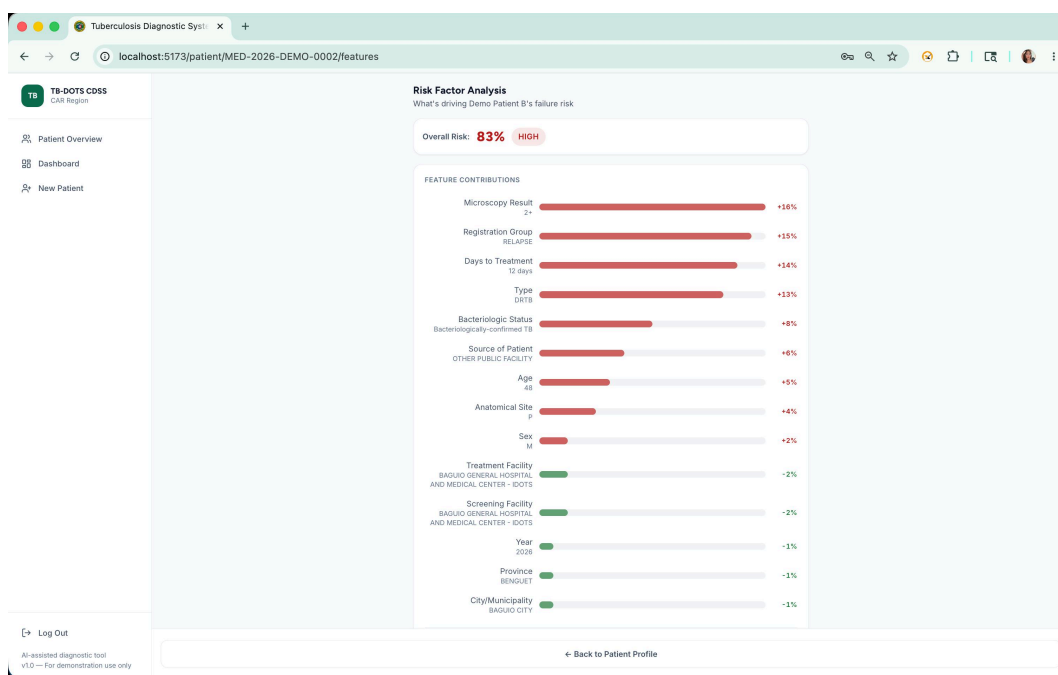


Figure 24*Initial Clinical Intervention and Management Protocol (First Step)*

The screenshot displays the 'TB-DOTS CDSS' web application interface. The browser address bar shows 'localhost:5173/patient/new'. The application has a sidebar with 'Patient Overview', 'Dashboard', and 'New Patient' (selected). A progress bar at the top indicates five steps: 1. Demographics, 2. Lab Results, 3. Chest X-rays, 4. Diagnosis, and 5. Treatment. The main content area is titled 'Patient Demographics' with a note: 'Basic patient information — all starred fields are required'. The form contains the following fields:

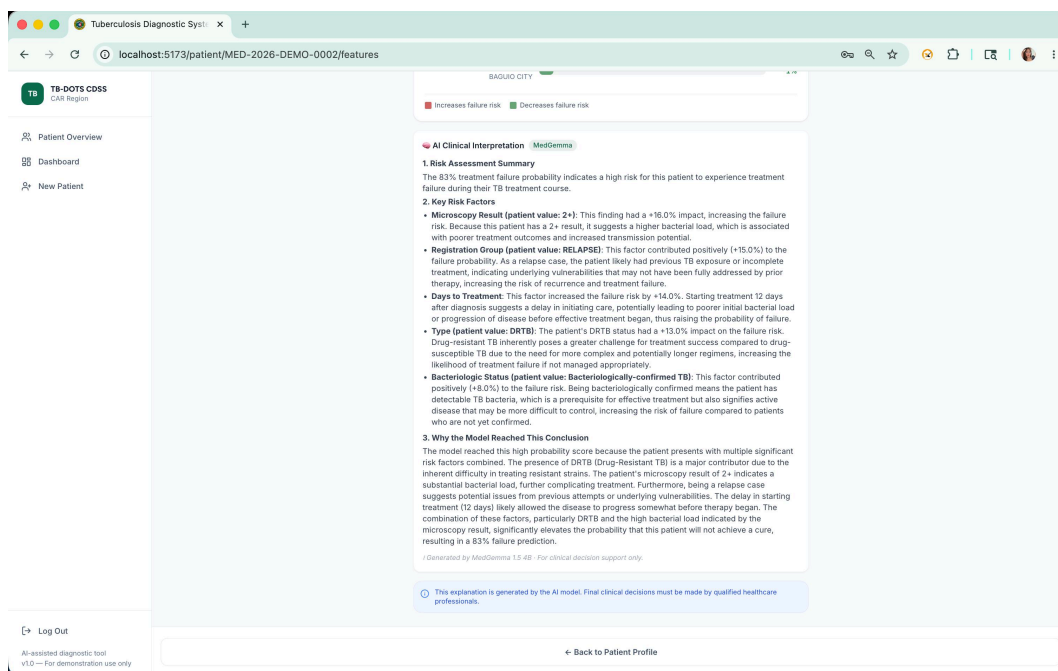
- Full Name ***: Text input with 'Juan dela Cruz' entered.
- Age ***: Text input with '35' entered.
- Gender ***: Dropdown menu with 'Select gender' selected.
- Medical ID**: Text input with 'MED-2026-7464' entered.
- Registration Group ***: Dropdown menu with 'Select group' selected.
- Province ***: Dropdown menu with 'Select province' selected.
- City / Municipality**: Dropdown menu with 'Select province first' selected.
- Civil Status**: Dropdown menu with 'Select Civil Status' selected.
- Nationality**: Text input with 'e.g. Filipino' entered.

At the bottom left, there is a 'Log Out' button and a footer note: 'AI-assisted diagnostic tool v1.0 — For demonstration use only'. At the bottom right, there is a green button labeled 'Continue to Lab Results'.

Large Language Model (LLM) Integration for Clinical Context

SHAP gives the quantitative feature importance; *Med-Gemma*, a fine-tuned medical large language model, adds a written account of the same risk factors. As Figure 25 shows, it sits inside the Risk Factor Analysis interface and reads the statistical weights back as plain clinical language.

Working from the specific patient profile (e.g., age, co-morbidities, and sputum conversion), *Med-Gemma* writes a short summary that puts the SHAP factor analysis into clinical terms. Instead of leaving the feature weights to stand on their own, it explains why the variables SHAP picked out matter for this patient. The result is that the CDSS reports more than a risk percentage; it also spells out the reasoning the model used, which makes the output easier for providers to read and check.

Figure 25*Med-Gemma Clinical Narrative and Risk Factor Explanation***Discussion**

This study looked at machine learning models for predicting treatment outcomes (Success or Failure) among PTB patients in the DOH-CAR TB-DOTS program, using monthly monitoring data together with baseline clinical features. Temporal modeling went through two iterations. The v1 iteration was an initial baseline, and its results show a common trap in imbalanced clinical prediction: a model can post high accuracy by siding with the majority Success class and still miss the real treatment failures. Its specificity was 0.0000 on the held-out test split, so Failure identification was not reliable under that first workflow.

The corrected v2 iteration improved the methodological defensibility of temporal prediction by addressing leakage risk, outcome-label handling, and temporal feature preparation. On the v2 held-out test split, the SMOTE-ENN tree models achieved consistent performance and identified 5 of 6 Failure cases while correctly classifying 36 of 38 Success cases. Random Forest had the strongest held-out ranking among the tree models (ROC-AUC 0.9693, PR-AUC 0.9950), while LightGBM reported the strongest cross-validated balanced accuracy in the saved artifacts. The v2 Hybrid Bi-LSTM baseline detected all Failure

cases on the held-out test split (specificity 1.0000) but raised more false Failure alerts among true Success cases, which lowered overall accuracy. This tradeoff can be acceptable in a screening-oriented CDSS where missing high-risk patients is more costly than additional review workload, but it must be managed to avoid over-alerting.

The study makes a few contributions. Pairing temporal monitoring data with static baseline features showed that patient-level longitudinal information predicts better than baseline data alone. Using SMOTE-ENN and the Bi-LSTM data augmentation against the extreme class imbalance in TB outcomes gave the models a better shot at catching high-risk patients. The work also shows why prediction has to come with explanation in a clinical setting: a risk estimate is easier to inspect once providers can see the patient-level factors behind the alert. SHAP explanations and bounded SLM summaries are that interpretability layer. They tie the model output to clinician review without taking the decision out of the clinician's hands.

The v2 temporal outputs look like a reasonable basis for further evaluation inside TB monitoring workflows, mainly for spotting earlier the patients who need closer review. For deployment, the v2 tree models (Random Forest and LightGBM with SMOTE-ENN especially) strike a good balance between Failure detection and overall correctness, while the Hybrid Bi-LSTM baseline suits a more cautious screening stance. Any clinical or programmatic use should first go through external validation, prospective testing, and review by healthcare professionals to check safety, calibration, and workflow fit. Future work should move to larger, multi-region datasets for better generalization, track temporal calibration over time, and test hybrids that combine deep temporal representations with tree-based learners. Adding interpretability over the temporal dynamics, such as attention across treatment months, may also help clinician trust and adoption.

Together these CDSS artifacts move the program from retrospective reporting toward earlier clinical management. NTP protocols usually catch treatment failure only after it happens; the predictive outputs instead allow intervention during the early months of therapy, when it still helps. The *Full Breakdown* report works against the black-box character of ensemble models, since it lets health workers check the model's logic against their own clinical experience. The *First Step* document then carries that accuracy into the care pathway, which can cut the MDR-TB that unrecognized non-adherence would otherwise produce. Pairing SHAP with Med-Gemma also speaks to the usual skepticism about AI in healthcare: with both a statistical breakdown and a written narrative, the CDSS keeps a human in the loop, letting DOH-CAR health workers weigh the model's findings against their own judgment before acting.

Recommendations

Healthcare facilities and policymakers should consider adding tree-based models (Random Forest and LightGBM with SMOTE-ENN especially) to the existing DOH-CAR TB-DOTS monitoring programs as a decision-support layer for early risk stratification. On the held-out test set these models detected treatment failure in a balanced way and still kept overall accuracy high, which makes them a practical way to flag patients who may need closer clinical review. Before any deployment they should be validated externally on independent multi-facility data, tested prospectively under real workflow conditions, and reviewed by healthcare professionals for calibration, safety, and fit with current practice.

The explainability layer warrants formal evaluation before any clinical deployment. A preliminary internal evaluation on the 44-patient test cohort found that the narrative layer consistently failed to echo the model's SHAP attribution when operating without access to the attribution signal: no cases passed the faithfulness criterion under the blind condition (0 of 44), regardless of the SLM backend used. Faithfulness improved substantially when SHAP contribution signals were included in the prompt, confirming that the narrative layer functions as a constrained verbalizer of the model's reasoning rather than an independent reasoner. Based on these findings, it is recommended that future work extend the faithfulness evaluation to a larger, multi-facility patient cohort, incorporate multiple SLM backends, and include clinician-rated readability, trust, and actionability as paired outcome measures to establish the evidence base needed for safe deployment of the explanation component.

Safety Scope and Interpretation Limits

The explanation layer is intended to support readability and interpretation of model reasoning by translating contribution signals into clinician-readable text, but its outputs must be interpreted within clear safety boundaries. SHAP explanations describe how features increased or decreased the model-predicted risk; they do not establish causal mechanisms of disease progression or prove that changing a highlighted factor will alter the patient's outcome. This distinction is important in high-stakes settings where generative outputs may otherwise be over-interpreted beyond their evidentiary basis .

Accordingly, the system is positioned as decision support: it flags risk, surfaces contributing factors,

and summarizes model behavior while preserving clinician authority for diagnosis and treatment decisions. This framing is consistent with contemporary reporting and governance guidance for AI-assisted clinical prediction systems, where transparency, bounded use, and human oversight remain central requirements .

The preliminary faithfulness evaluation described above supports this bounded framing by showing that the narrative layer depends on SHAP contribution signals rather than reconstructing model attribution from patient context alone. Accordingly, the operational design retains SHAP as the primary explanation surface and treats the narrative output as a readability aid rather than a substitute for the attribution itself.

Conclusion

Tuberculosis treatment in the Cordillera Administrative Region (CAR) is shaped by very different patient situations, gaps in record-keeping, and disruptions to the health program. These are problems that simple one-variable analyses cannot fully explain. To turn this raw data into something useful for prediction, the study first built a clear and repeatable cleaning pipeline for two related datasets: a 7,389-patient DOH-CAR registry and a 599-patient month-by-month monitoring set. The pipeline standardized formats, removed patient identifiers, and handled missing values by first checking why each variable was missing (MCAR, MAR, or MNAR) before applying one of four matching strategies. Because the cleaned data was trustworthy rather than blindly filled in, the descriptive and exploratory analyses could clearly show outcome trends over the years, the rise in deaths and lost follow-ups during the COVID-19 period, and the patient-level, time-based, and operational patterns that later guided model design.

These cleaned signals fed directly into how well the models performed. On the static dataset, XGBoost with SMOTE-ENN gave the strongest screening-focused setup (composite score $S = 0.699$; failure recall 72.9%), while XGBoost without resampling produced the best-calibrated risk ranking (test AUC 0.712; MCC 0.258), offering two different setups depending on how the tool will be used. The longitudinal models pushed performance further: after the v2 pipeline fixed the data leakage and label-handling issues that had limited v1, the SMOTE-ENN tree models (Random Forest, LightGBM, and XGBoost) reached 0.9318 accuracy and 0.8333 specificity, correctly catching five of six failure cases in the held-out test set, while the Hybrid Bi-LSTM caught every failure case but produced more false alarms. Building on these predictions, SHAP-based feature explanations and a bounded SLM narrative layer were combined into a

Clinical Decision Support System (CDSS) that shows each patient's risk level, the factors driving that risk, and updated estimates as new monthly data come in.

These results are strong enough to be clinically promising, but not yet ready for actual clinical use. Since the test cases came from only one regional registry, and since the narrative layer's faithfulness was only checked on a small group of 44 patients, the system must first go through external validation using data from other facilities, prospective testing in real clinical workflows, and review by healthcare professionals to confirm safety, calibration, and fit. The narrative explanation layer will also need a larger faithfulness study before it can be safely used in clinical practice. Once these conditions are met, the framework and the published pseudonymized version of the DOH-CAR PTB Treatment Outcome Dataset may only then be able to support both regional TB control efforts and future research on explainable clinical prediction.

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